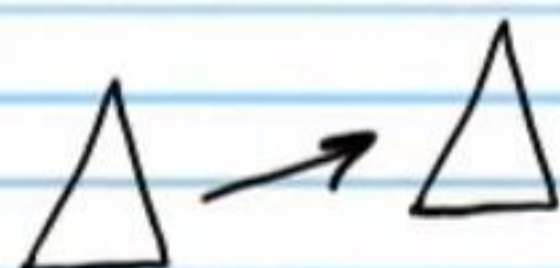


## Transformations and the Library of Relations (Part II)

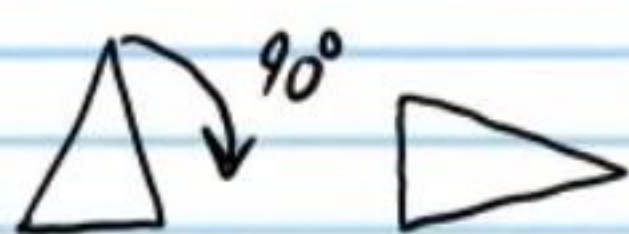
### Review on Transformations

#### I) Translation.



#### II) Rotation.

\* There must be a center of rotation



#### III) Reflection.

\* There must be a line of reflection.



#### IV) Dilation.

\* There must be a center of dilation and a scale factor.



No Change  
in Size.

**Rigid**

Size Changes.

**Non-Rigid**

## Relations

**Relation:** A connection between two values (usually called  $x$  and  $y$ ).

### Three Ways to Represent a Relation

#### Examples

#### I. Equation

$$y = x^2$$

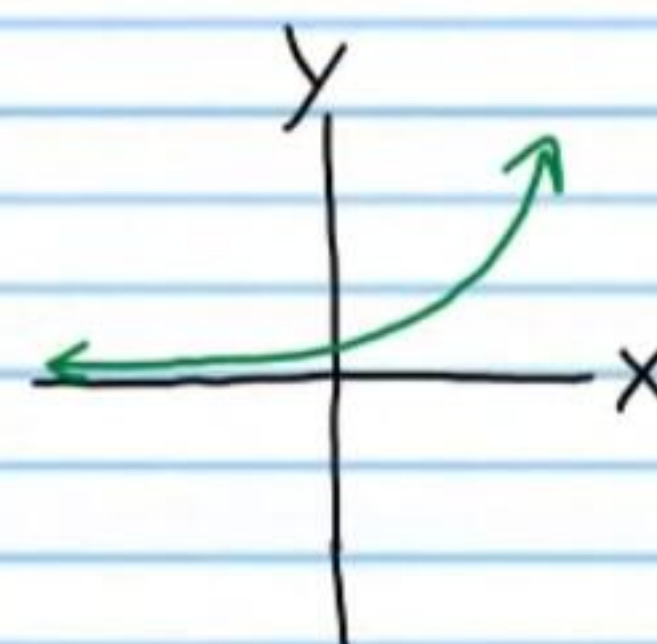
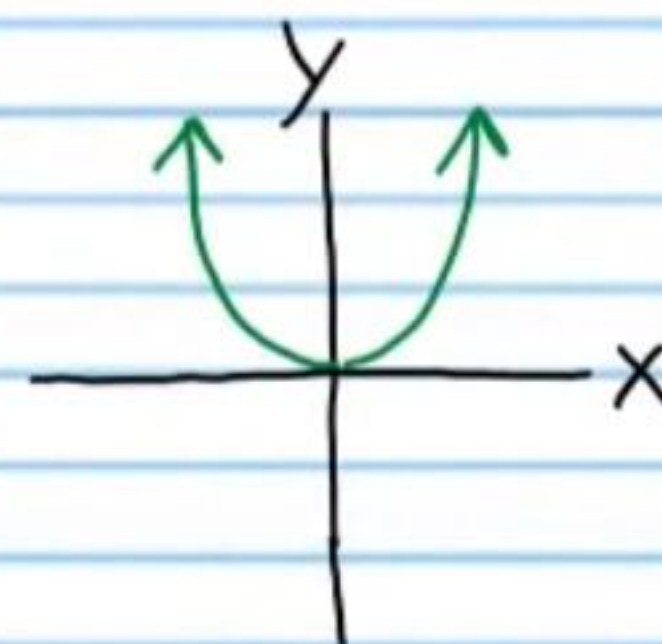
$$y = 3^x$$

#### II. Table

|     |    |    |   |   |   |
|-----|----|----|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 |
| $y$ | 4  | 1  | 0 | 1 | 4 |

|     |               |               |   |   |   |
|-----|---------------|---------------|---|---|---|
| $x$ | -2            | -1            | 0 | 1 | 2 |
| $y$ | $\frac{1}{9}$ | $\frac{1}{3}$ | 1 | 3 | 9 |

#### III. Graph

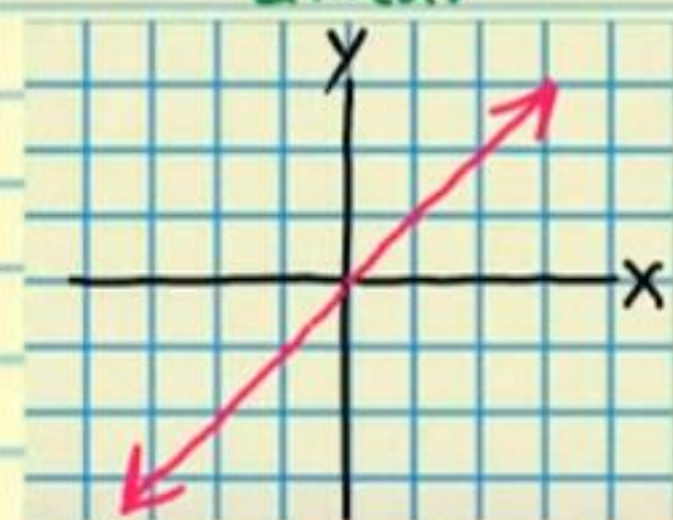




# The Library of Relations

\* Also called "parent" relations.

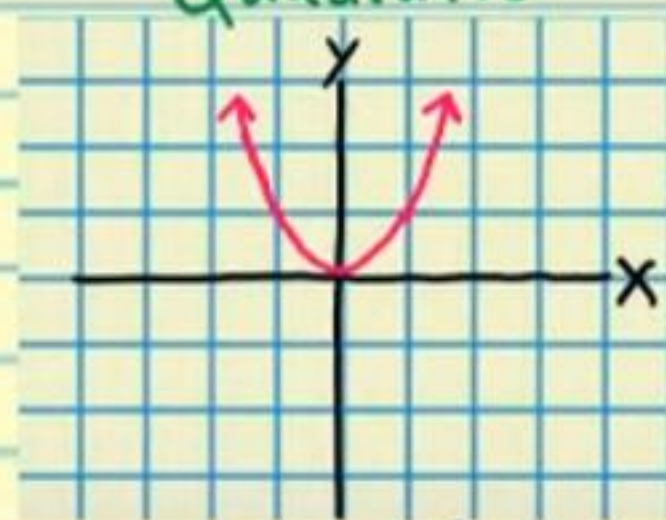
Linear



$$Y = X$$

|   |    |    |   |   |   |
|---|----|----|---|---|---|
| X | -2 | -1 | 0 | 1 | 2 |
| Y | -2 | -1 | 0 | 1 | 2 |

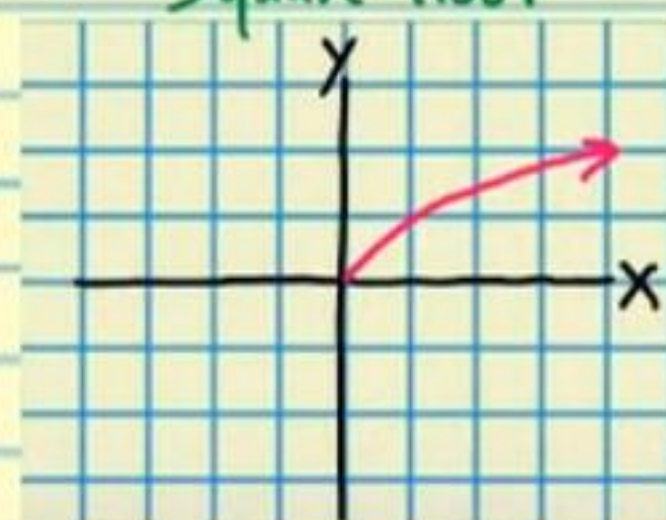
Quadratic



$$Y = x^2$$

|   |    |    |   |   |   |
|---|----|----|---|---|---|
| X | -2 | -1 | 0 | 1 | 2 |
| Y | 4  | 1  | 0 | 1 | 4 |

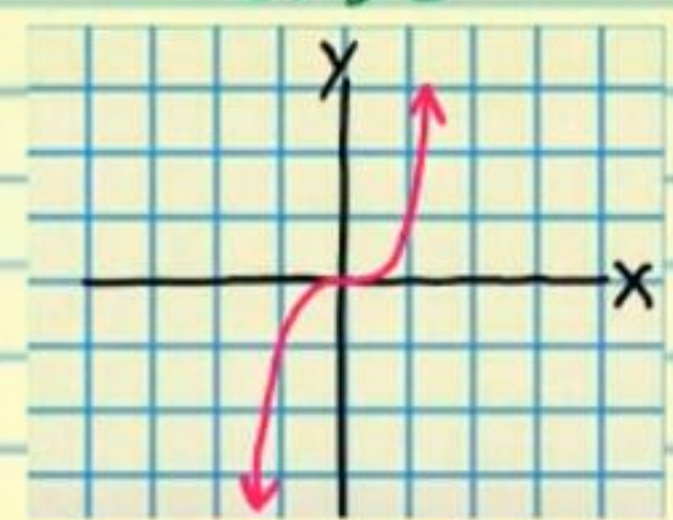
Square Root



$$Y = \sqrt{X}$$

|   |   |   |   |   |    |
|---|---|---|---|---|----|
| X | 0 | 1 | 4 | 9 | 16 |
| Y | 0 | 1 | 2 | 3 | 4  |

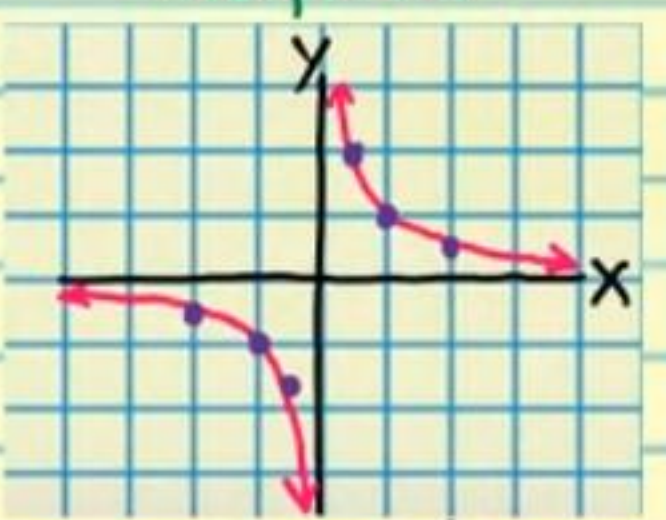
Cubic



$$Y = x^3$$

|   |    |    |   |   |   |
|---|----|----|---|---|---|
| X | -2 | -1 | 0 | 1 | 2 |
| Y | -8 | -1 | 0 | 1 | 8 |

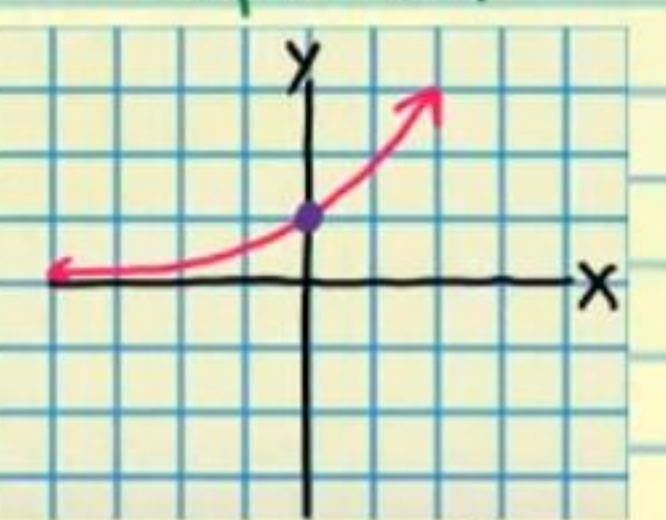
Reciprocal



$$Y = \frac{1}{x}$$

|   |      |    |      |           |     |   |     |
|---|------|----|------|-----------|-----|---|-----|
| X | -2   | -1 | -1/2 | 0         | 1/2 | 1 | 2   |
| Y | -1/2 | -1 | -2   | undefined | 2   | 1 | 1/2 |

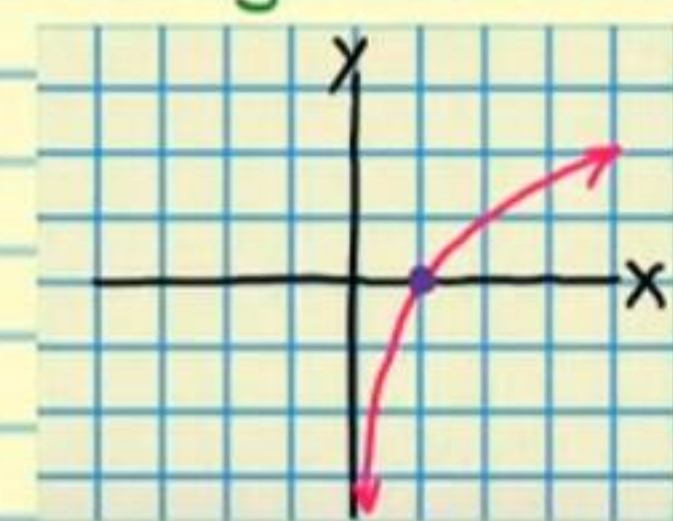
Exponential



$$Y = a^x, a > 1$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| X | - | - | 0 | - | - |
| Y | - | - | 1 | - | - |

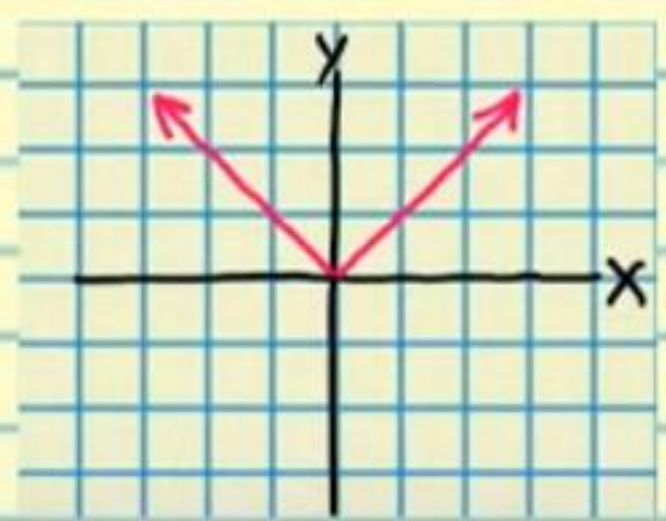
Logarithmic



$$Y = \log_a X$$

|   |           |   |   |   |   |
|---|-----------|---|---|---|---|
| X | 0         | 1 | - | - | - |
| Y | undefined | 0 | - | - | - |

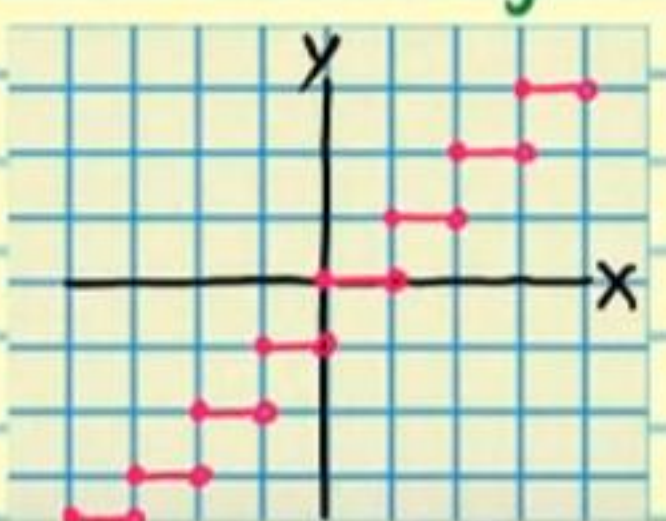
Absolute Value



$$Y = |x|$$

|   |    |    |   |   |   |
|---|----|----|---|---|---|
| X | -2 | -1 | 0 | 1 | 2 |
| Y | 2  | 1  | 0 | 1 | 2 |

Greatest Integer



$$Y = [x]$$

|   |        |    |   |     |       |
|---|--------|----|---|-----|-------|
| X | -2 1/2 | -1 | 0 | 1/2 | 1 1/2 |
| Y | -3     | -1 | 0 | 0   | 1     |

# The Transformation Properties

## Translation

If  $y$  is replaced with  $y+k$  in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved  $k$  units **down** (assuming that  $k$  is positive).

If  $y$  is replaced with  $y-k$  in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved  $k$  units **up** (assuming that  $k$  is positive).

If  $x$  is replaced with  $x+k$  in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved  $k$  units to the **left** (assuming that  $k$  is positive).

If  $x$  is replaced with  $x-k$  in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved  $k$  units to the **right** (assuming that  $k$  is positive).

## Reflection

If  $y$  is replaced with  $-y$  in an equation, then the graph of the new equation is the same as the graph of the original equation, but reflected across the **x-axis**.

If  $x$  is replaced with  $-x$  in an equation, then the graph of



the new equation is the same as the graph of the original equation, but reflected across the  $y$ -axis.

## Dilation

### Vertical Compression

If  $y$  is replaced with  $Ky$  in an equation, then the graph of the new equation is the same as the graph of the original equation, except every  $y$ -value is **divided** by  $K$  (assuming that  $K$  is positive).

### Vertical Stretch

If  $y$  is replaced with  $y/K$  in an equation, then the graph of the new equation is the same as the graph of the original equation, except every  $y$ -value is **multiplied** by  $K$  (assuming that  $K$  is positive).

### Horizontal Compression

If  $x$  is replaced with  $Kx$  in an equation, then the graph of the new equation is the same as the graph of the original equation, except every  $x$ -value is **divided** by  $K$  (assuming that  $K$  is positive).

### Horizontal Stretch

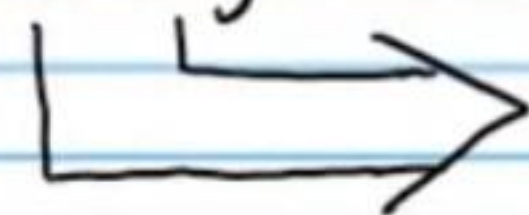
If  $x$  is replaced with  $x/K$  in an equation, then the graph of the new equation is the same as the graph of the original equation, except every  $x$ -value is **multiplied** by  $K$  (assuming that  $K$  is positive).

## Altering Equations to Draw Their Graphs

Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

①  $y = 2x^3 + 1$

Change the form of the equation.



$$y = 2x^3 + 1$$

$$y - 1 = 2x^3 + 1 - 1$$

$$y - 1 = 2x^3$$

$$\frac{y - 1}{2} = \frac{2x^3}{2}$$

Altered Form:  $\frac{y - 1}{2} = x^3$

Parent Relation

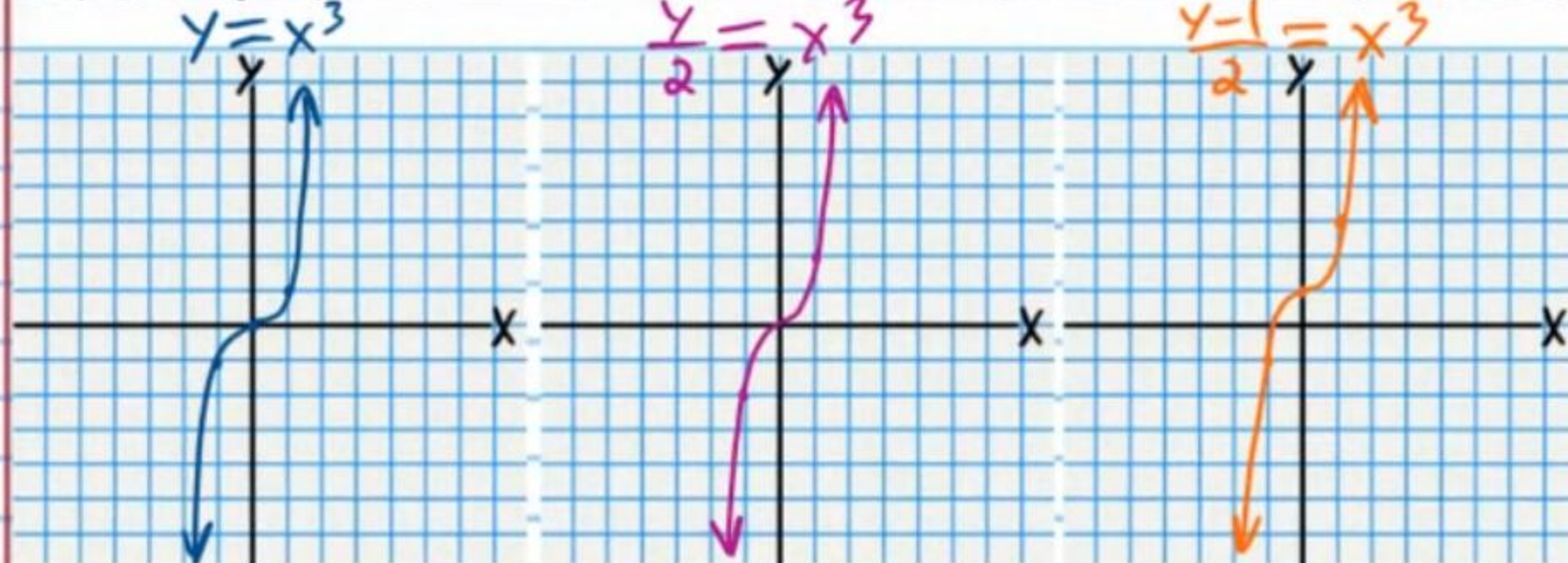
$$y = x^3$$

Transformation #1

$$\frac{y}{2} = x^3$$

Transformation #2

$$\frac{y - 1}{2} = x^3$$





$$\textcircled{2} y = \frac{\log_2 X}{3} - 4$$

Change the form of the equation.

$$\Rightarrow y = \frac{\log_2 X}{3} - 4$$

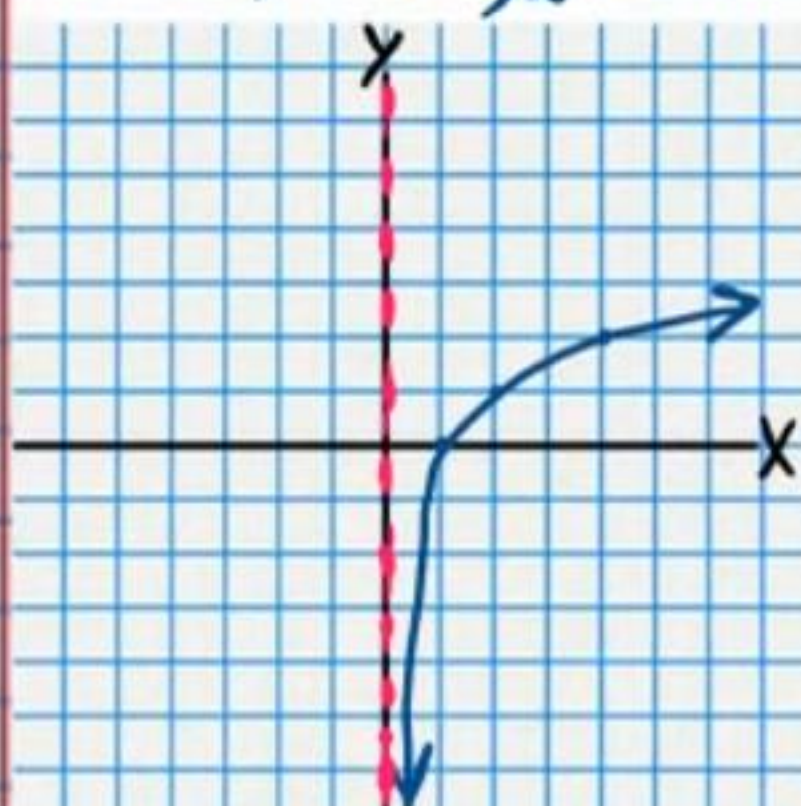
$$y + 4 = \frac{\log_2 X}{3} - 4 + 4$$

$$y + 4 = \frac{\log_2 X}{3}$$

$$3(y + 4) = \log_2 X \quad (3)$$

Altered Form:  $3(y + 4) = \log_2 X$

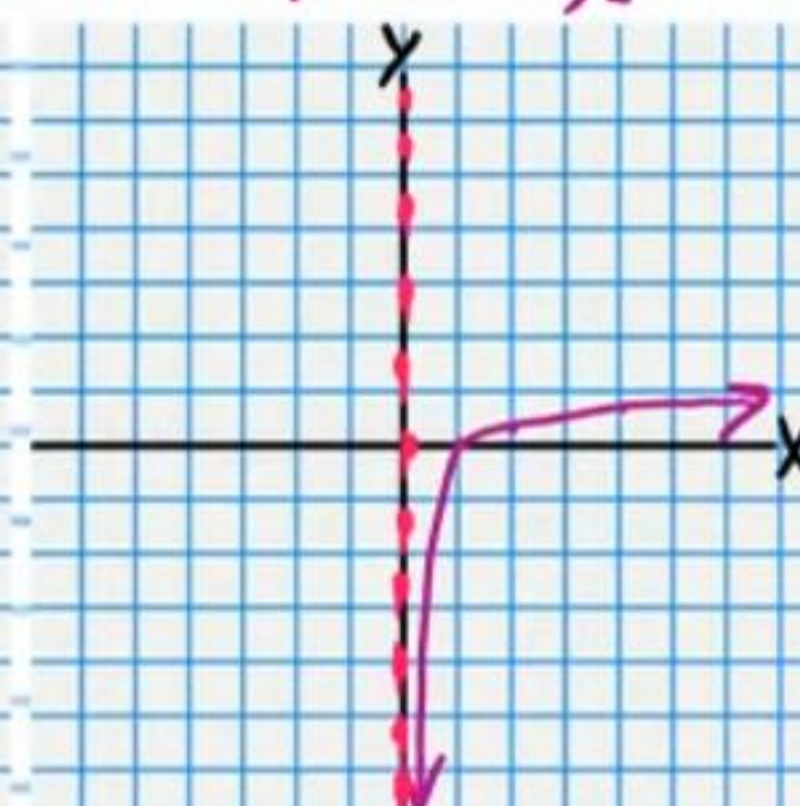
Parent Relation  
 $y = \log_2 X$



Logarithm Property III

$$\log_b b = 1$$

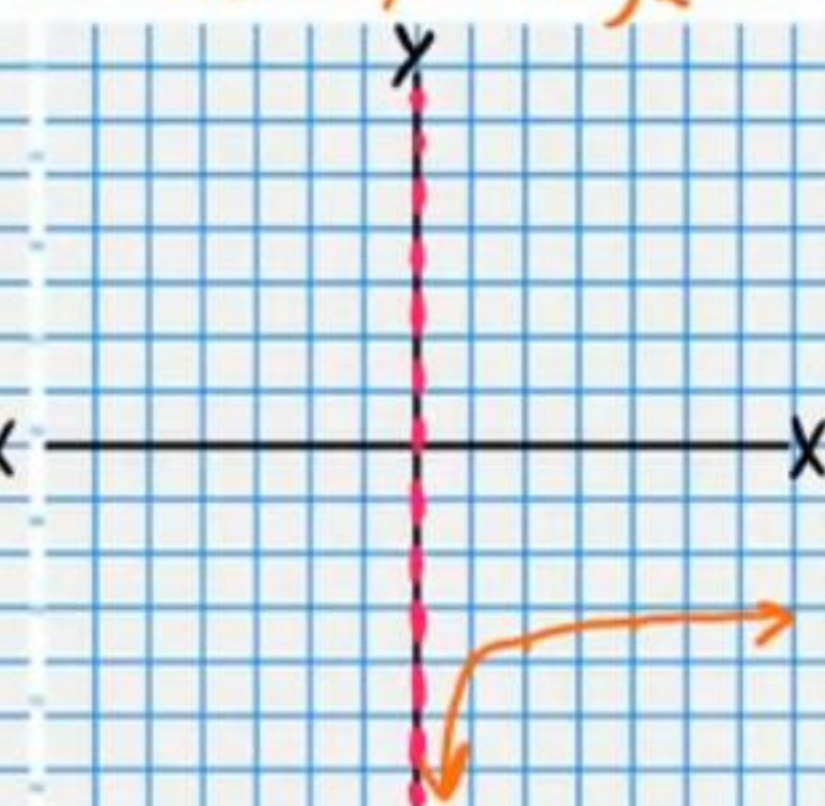
Transformation #1  
 $3y = \log_2 X$



Logarithm Property II

$$\log_b b^x = x$$

Transformation #2  
 $3(y + 4) = \log_2 X$



Logarithm Property IX

$$\log_b 1 = 0$$

$$\textcircled{3} y = 4|x| + 3$$

Change the form of the equation.

$$\Rightarrow y = 4|x| + 3$$

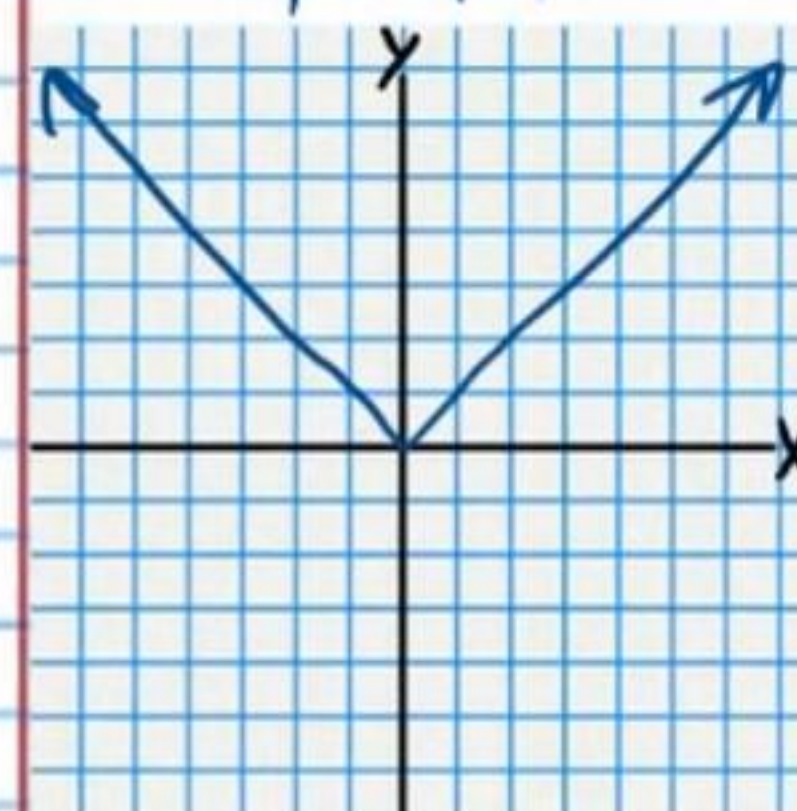
$$y - 3 = 4|x| + 3 - 3$$

$$y - 3 = 4|x|$$

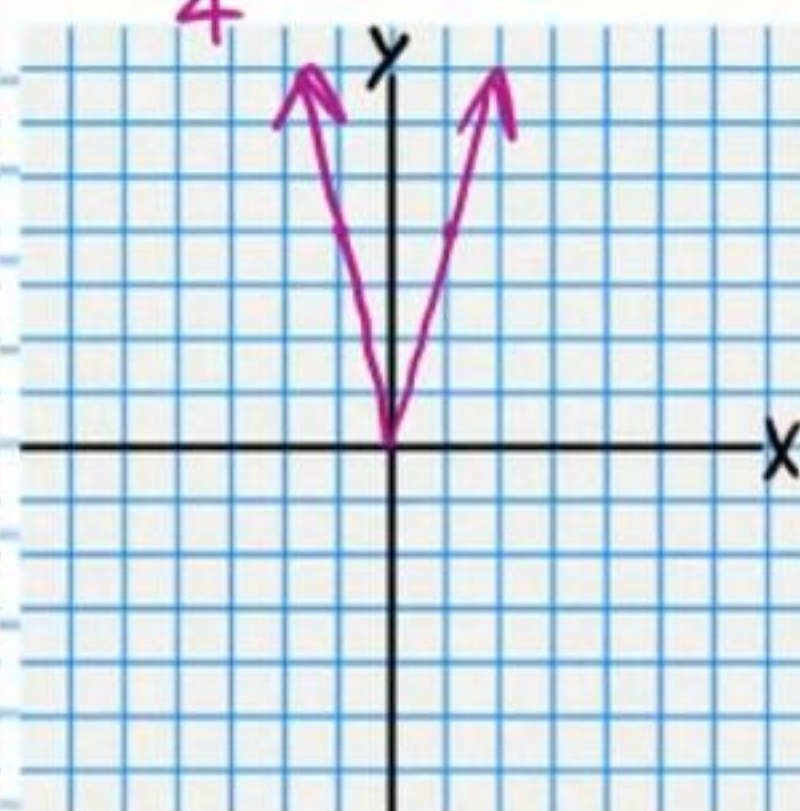
$$\frac{y - 3}{4} = \frac{4|x|}{4}$$

Altered Form:  $\frac{y - 3}{4} = |x|$

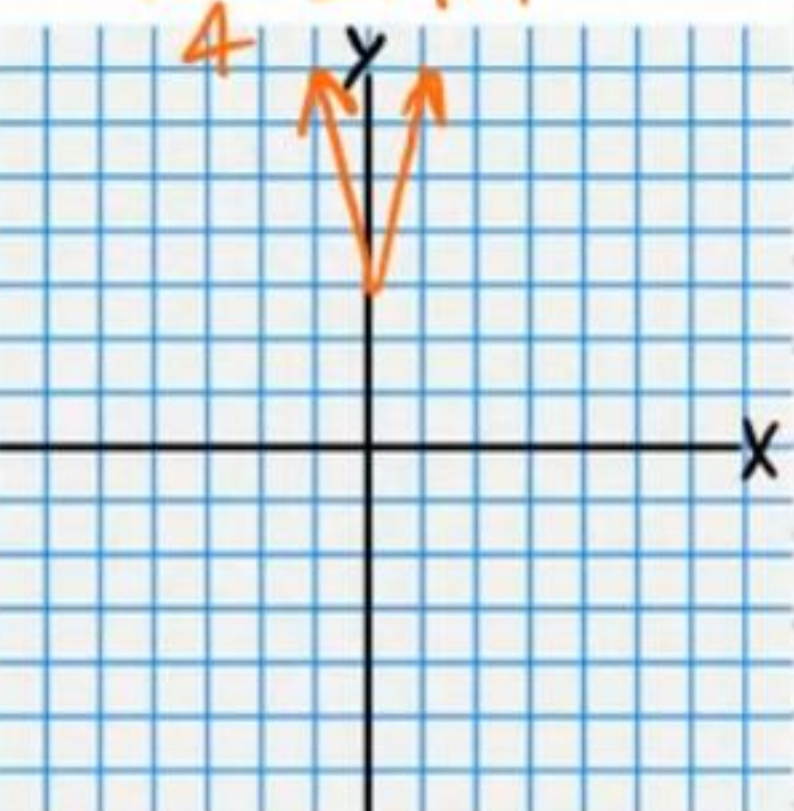
Parent Relation  
 $y = |x|$



Transformation #1  
 $\frac{y}{4} = |x|$



Transformation #2  
 $\frac{y - 3}{4} = |x|$





$$\textcircled{4} y = \frac{3^x}{4} - 1$$

Change the form of the equation.

$$\Rightarrow y = \frac{3^x}{4} - 1$$

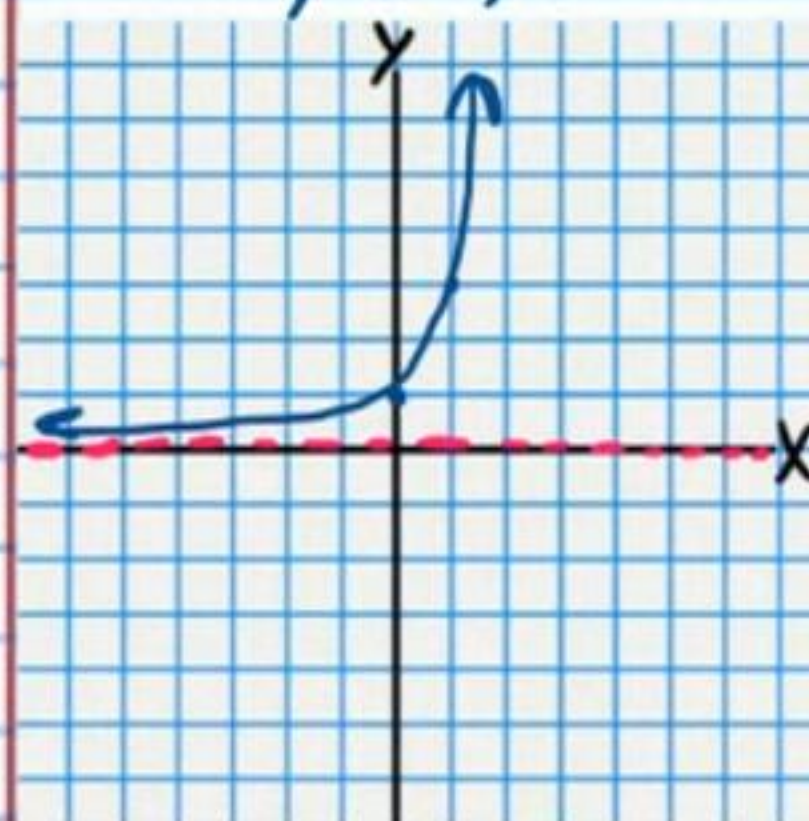
$$y + 1 = \frac{3^x}{4} - 1 + 1$$

$$y + 1 = \frac{3^x}{4}$$

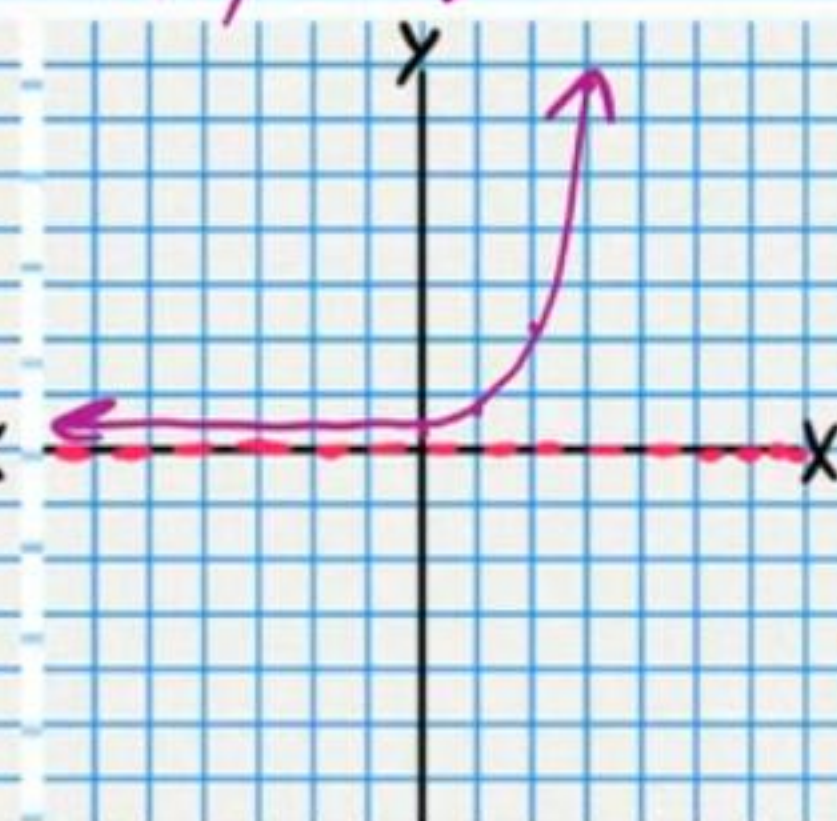
$$4(y + 1) = \frac{3^x}{4} (4)$$

Altered Form:  $4(y + 1) = 3^x$

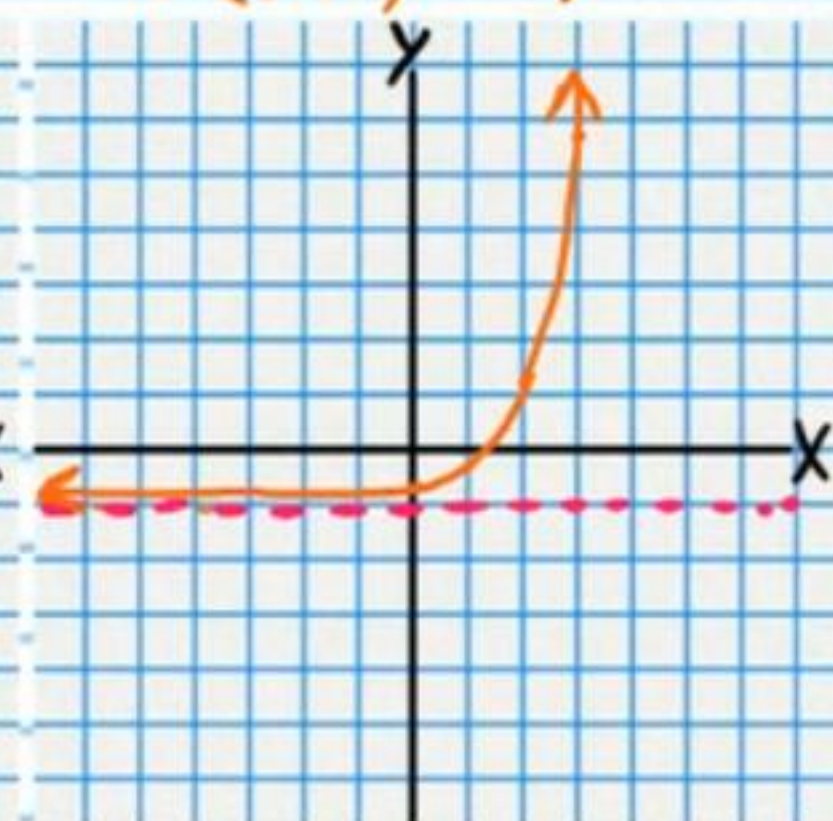
Parent Relation  
 $y = 3^x$



Transformation #1  
 $4y = 3^x$



Transformation #2  
 $4(y + 1) = 3^x$



$$\textcircled{5} y = -\sqrt{x} + 6$$

Change the form of the equation.

$$\Rightarrow y = -\sqrt{x} + 6$$

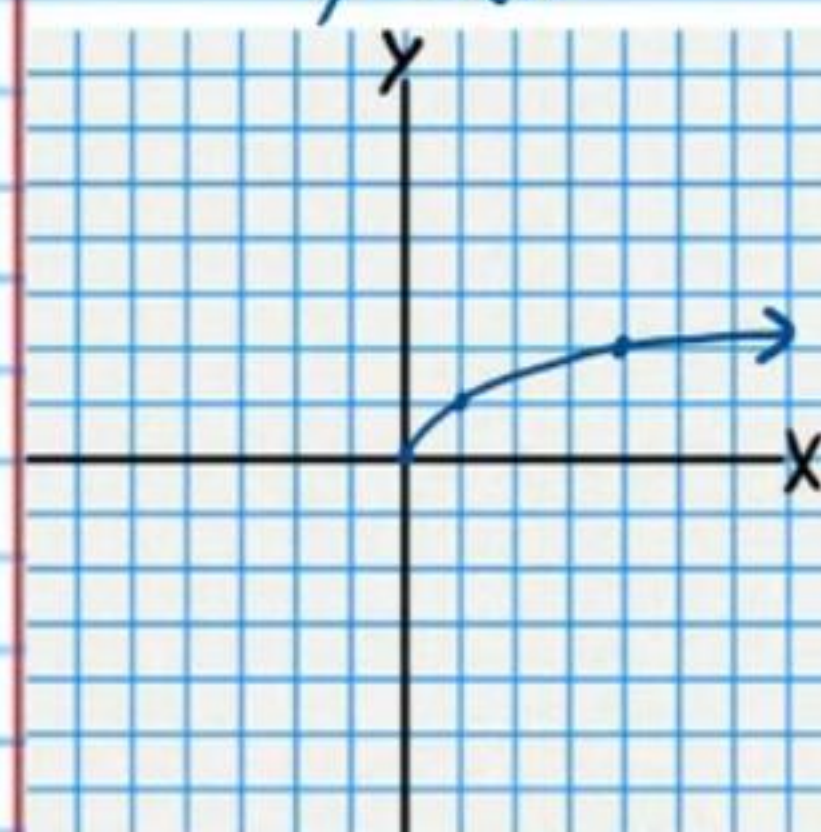
$$y - 6 = -\sqrt{x} + 6 - 6$$

$$y - 6 = -\sqrt{x}$$

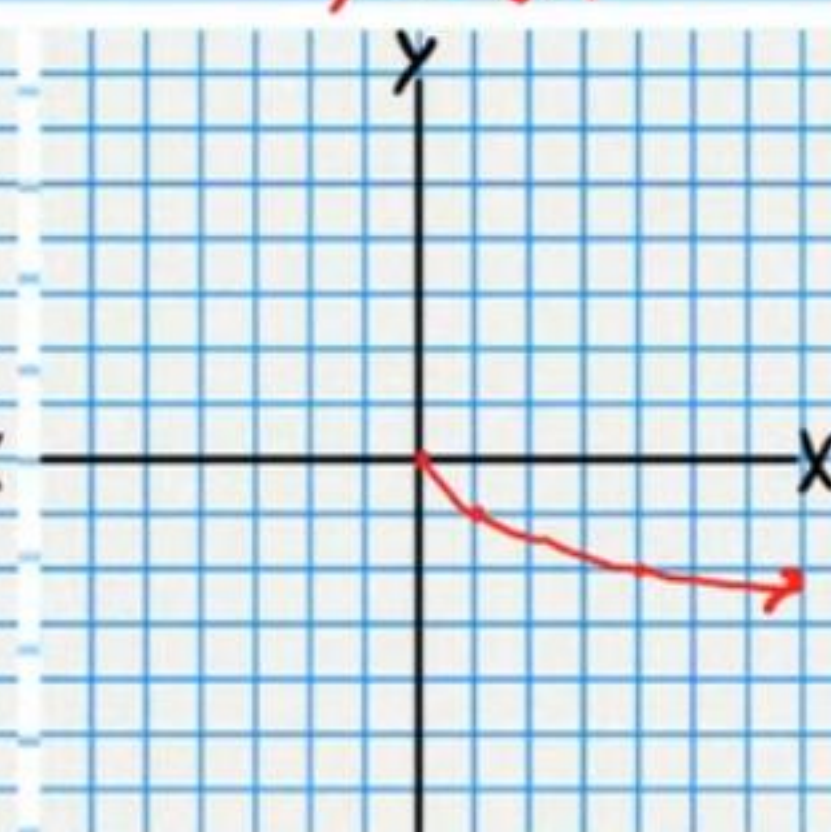
$$\underline{y - 6} = \underline{-\sqrt{x}}$$

Altered Form:  $-(y - 6) = \sqrt{x}$

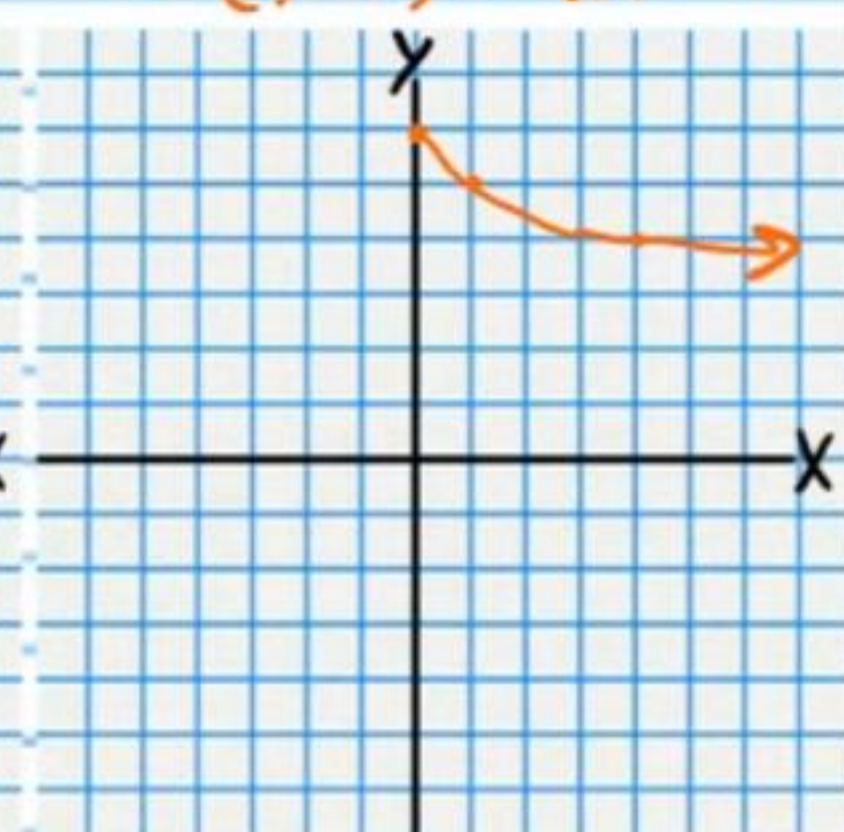
Parent Relation  
 $y = \sqrt{x}$



Transformation #1  
 $-y = \sqrt{x}$



Transformation #2  
 $-(y - 6) = \sqrt{x}$





$$\textcircled{6} y = -\frac{1}{x} - 3$$

Change the form of the equation.

$$\Rightarrow y = -\frac{1}{x} - 3$$

$$y + 3 = -\frac{1}{x} - 3 + 3$$

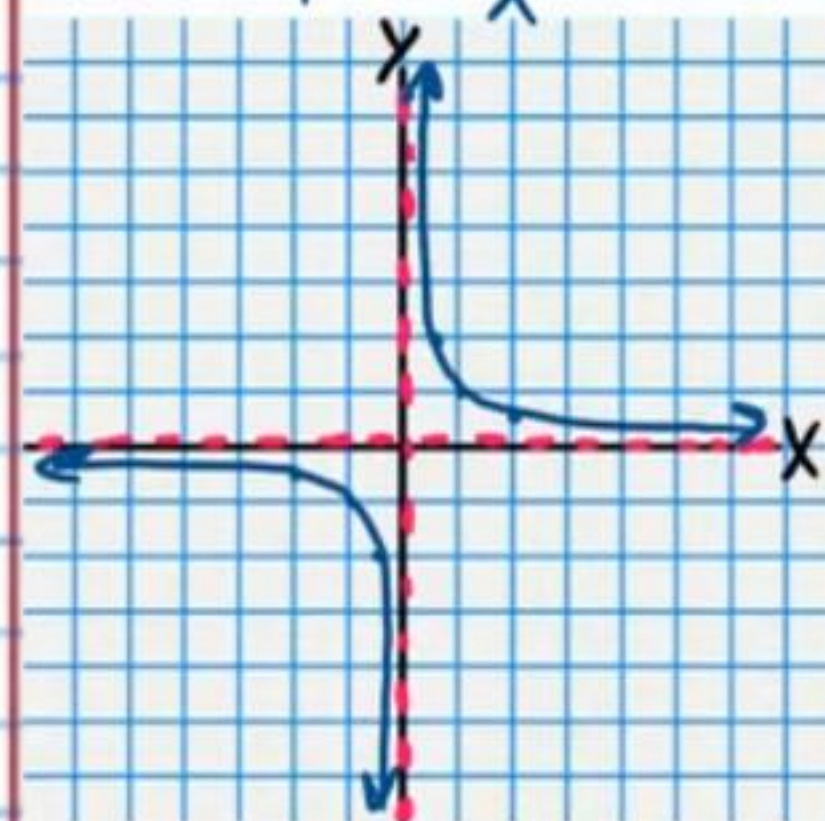
$$y + 3 = -\frac{1}{x}$$

$$-1(y + 3) = -\frac{1}{x}(-1)$$

$$\text{Altered Form: } -(y + 3) = \frac{1}{x}$$

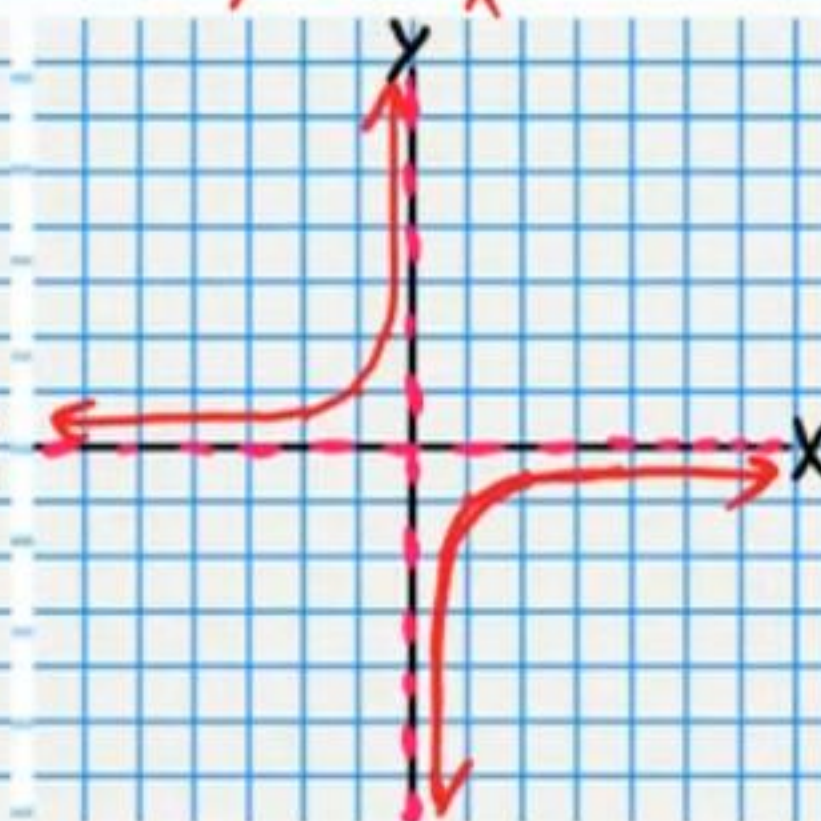
Parent Relation

$$y = \frac{1}{x}$$



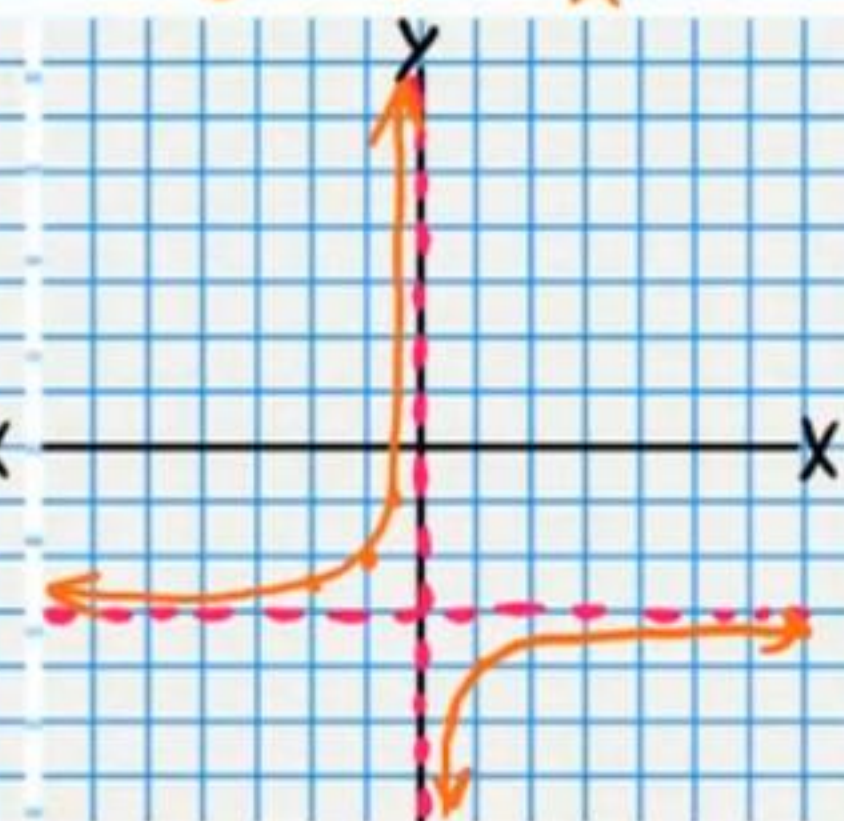
Transformation #1

$$-y = \frac{1}{x}$$



Transformation #2

$$-(y + 3) = \frac{1}{x}$$



$$\textcircled{7} y = -x^2 + 5$$

Change the form of the equation.

$$\Rightarrow y = -x^2 + 5$$

$$y - 5 = -x^2 + 5 - 5$$

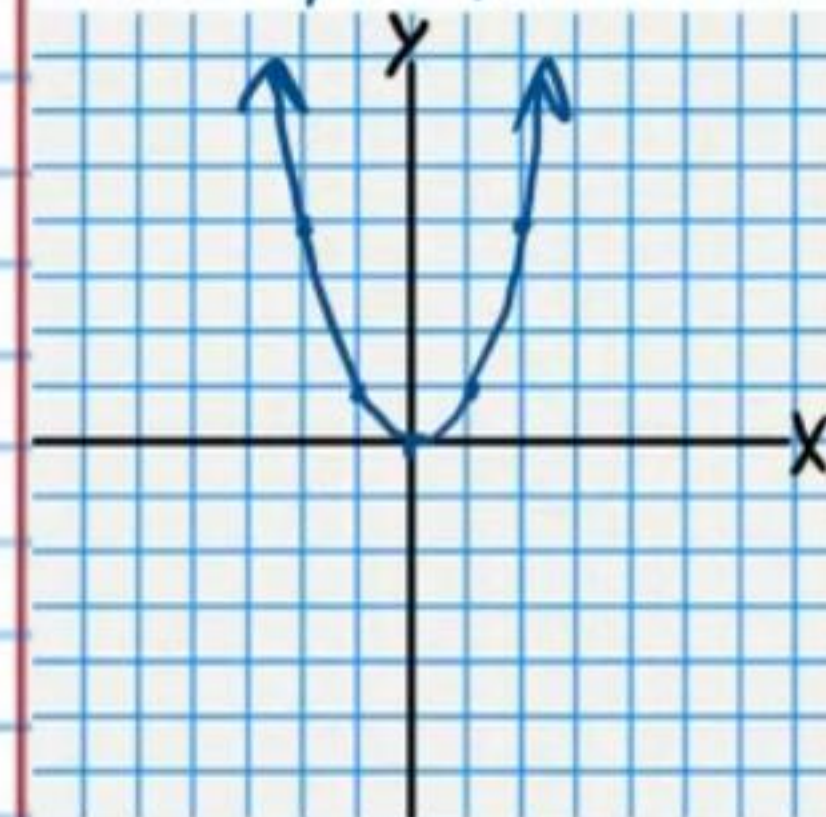
$$y - 5 = -x^2$$

$$\frac{y - 5}{-1} = \frac{-x^2}{-1}$$

$$\text{Altered Form: } -(y - 5) = x^2$$

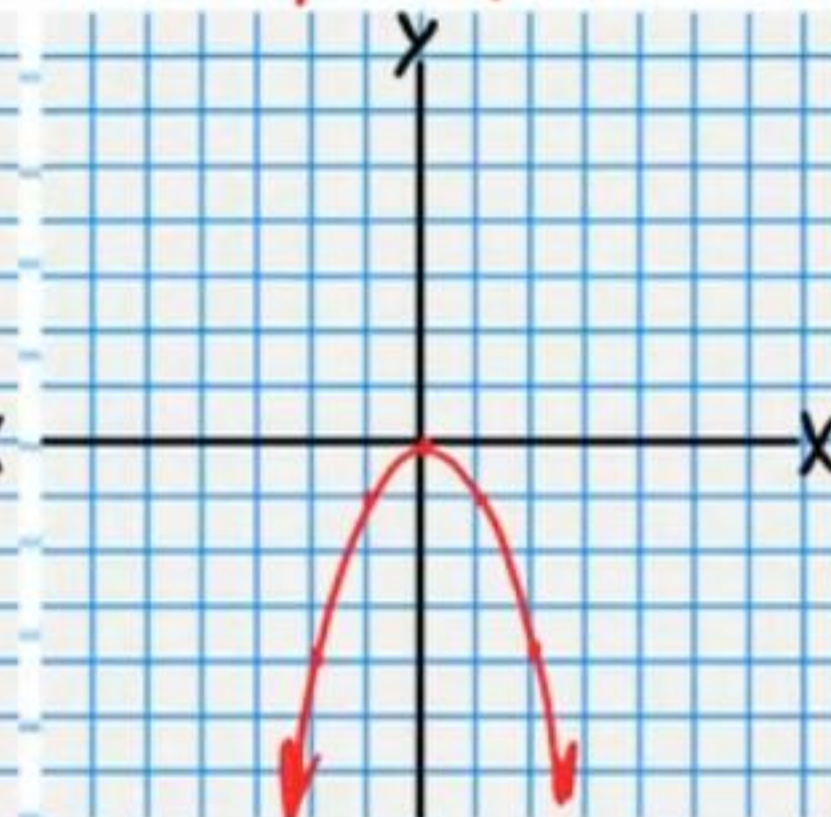
Parent Relation

$$y = x^2$$



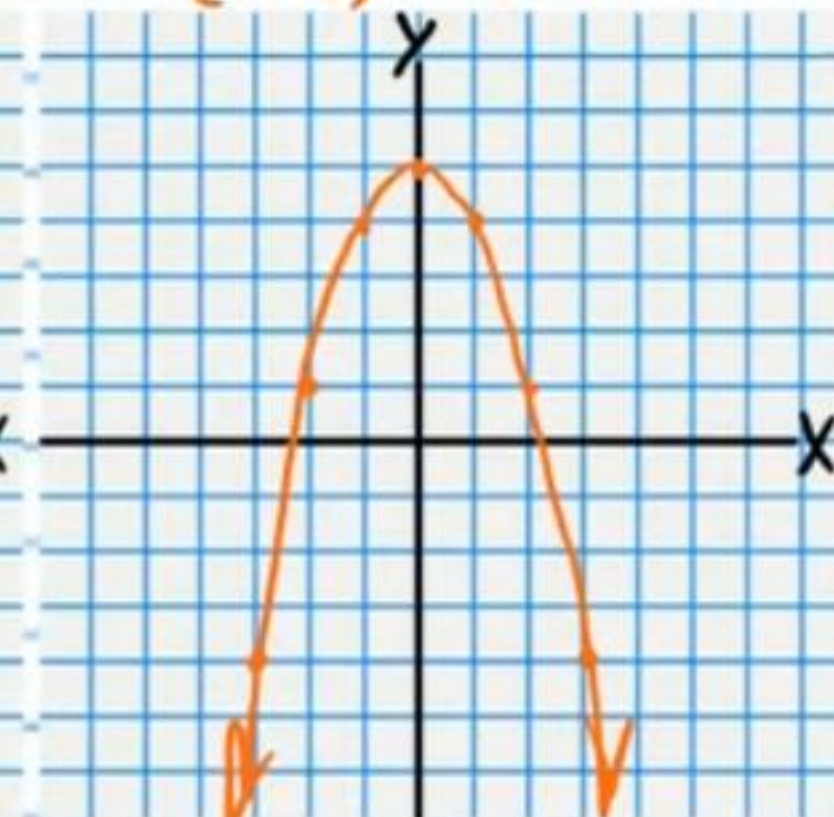
Transformation #1

$$-y = x^2$$



Transformation #2

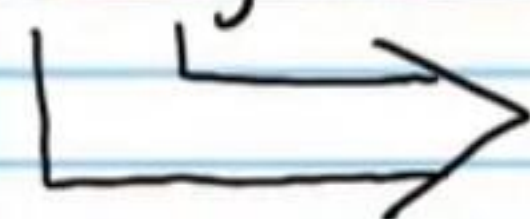
$$-(y - 5) = x^2$$





$$\textcircled{8} y = -2^x - 4$$

Change the form of the equation.



$$y = -2^x - 4$$

$$y + 4 = -2^x - 4 + 4$$

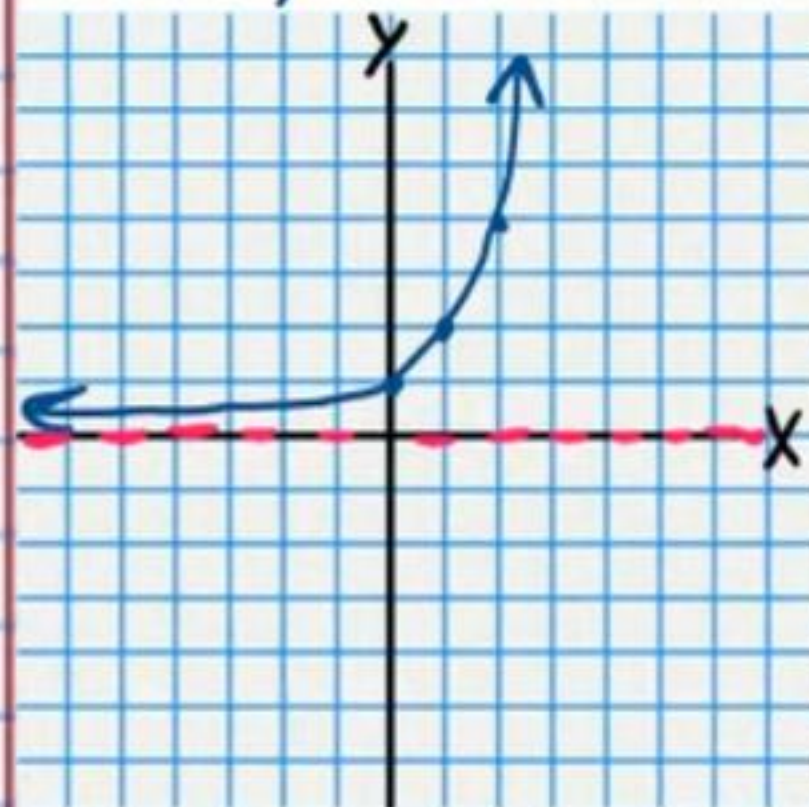
$$y + 4 = -2^x$$

$$\frac{y+4}{-1} = \frac{-2^x}{-1}$$

Altered Form:  $-(y+4) = 2^x$

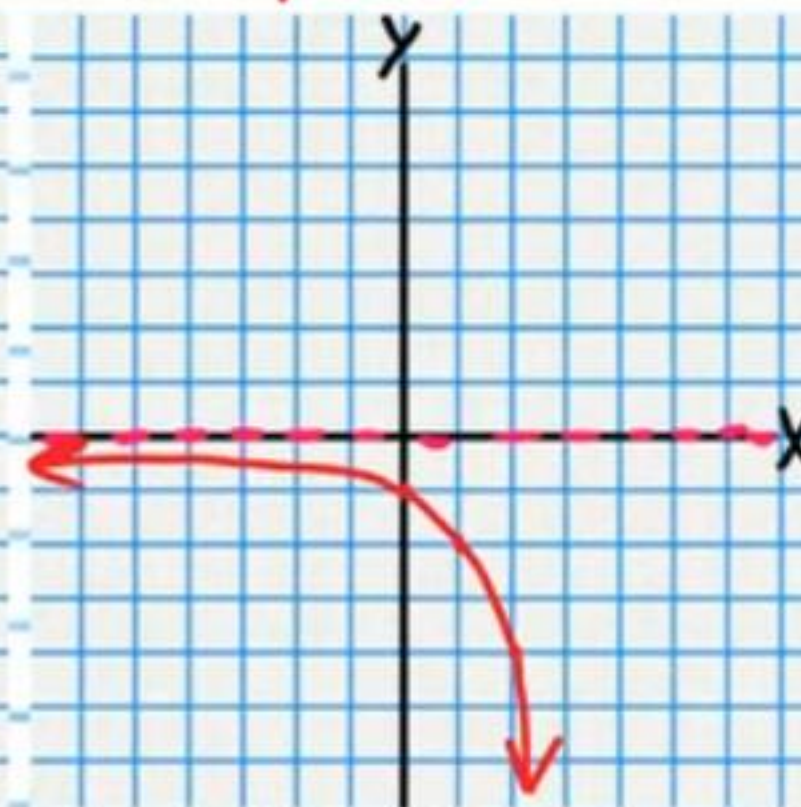
Parent Relation

$$y = 2^x$$



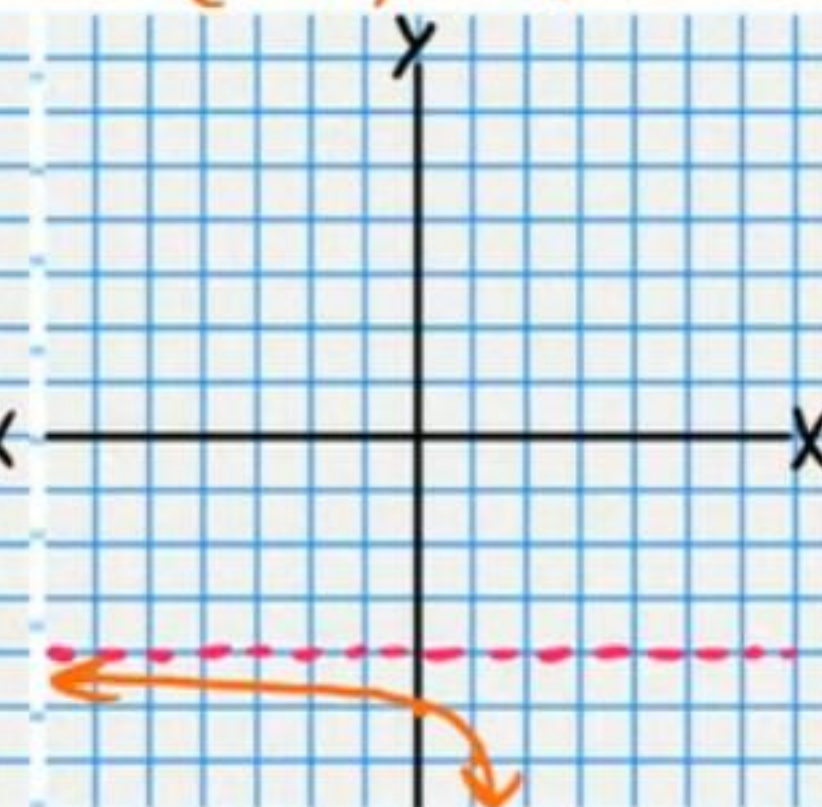
Transformation #1

$$-y = 2^x$$



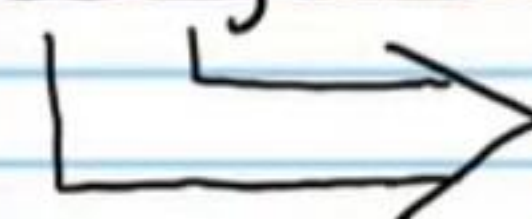
Transformation #2

$$-(y+4) = 2^x$$



$$\textcircled{9} y = 3\log_4(x+1) - 2$$

Change the form of the equation.



$$y = 3\log_4(x+1) - 2$$

$$y + 2 = 3\log_4(x+1) - 2 + 2$$

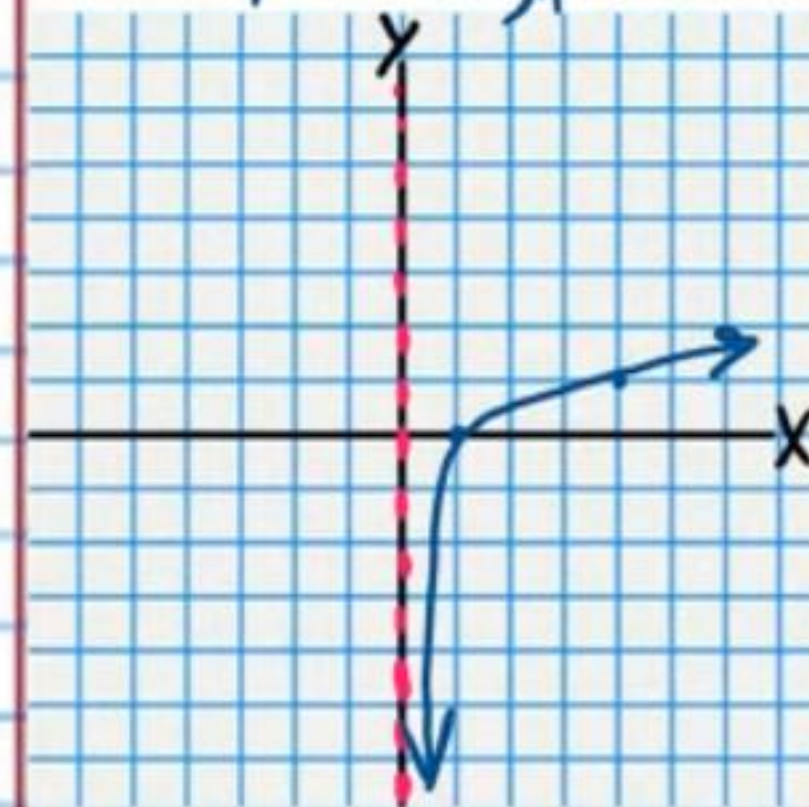
$$y + 2 = 3\log_4(x+1)$$

$$\frac{y+2}{3} = \frac{3\log_4(x+1)}{3}$$

Altered Form:  $\frac{y+2}{3} = \log_4(x+1)$

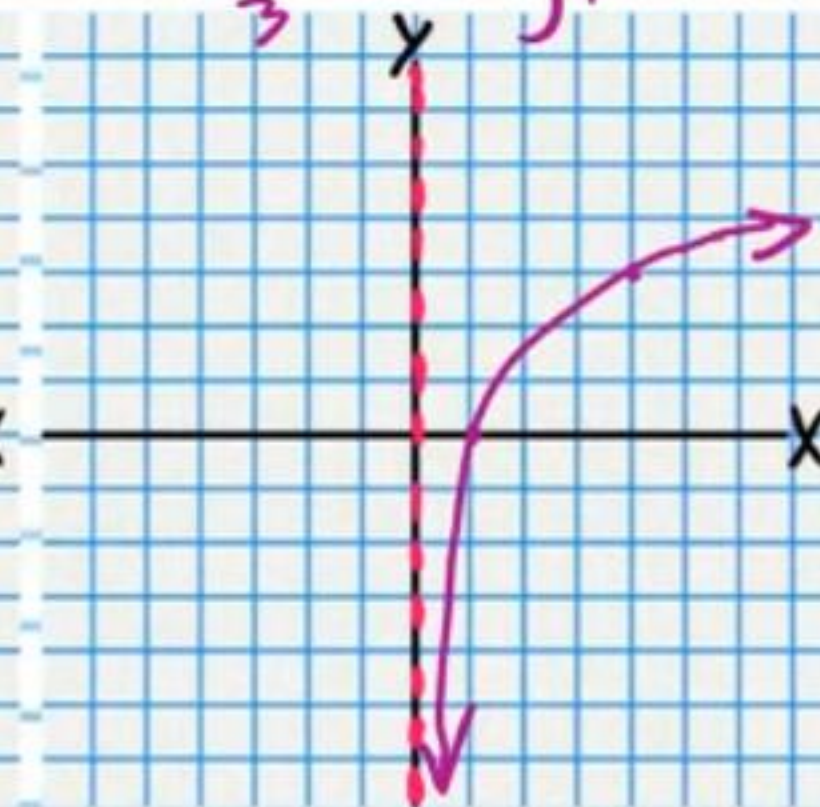
Parent Relation

$$y = \log_4 x$$



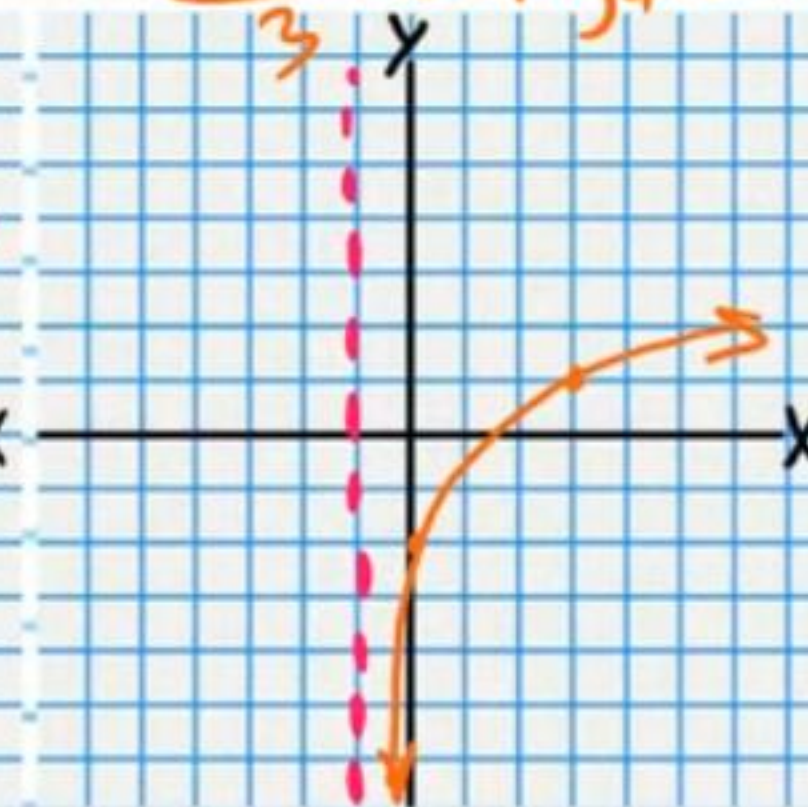
Transformation #1

$$\frac{y}{3} = \log_4 x$$



Transformation #2

$$\frac{y+2}{3} = \log_4(x+1)$$



Logarithm Property III

$$\log_b b = 1$$

Logarithm Property II

$$\log_b b^x = x$$

Logarithm Property IX

$$\log_b 1 = 0$$



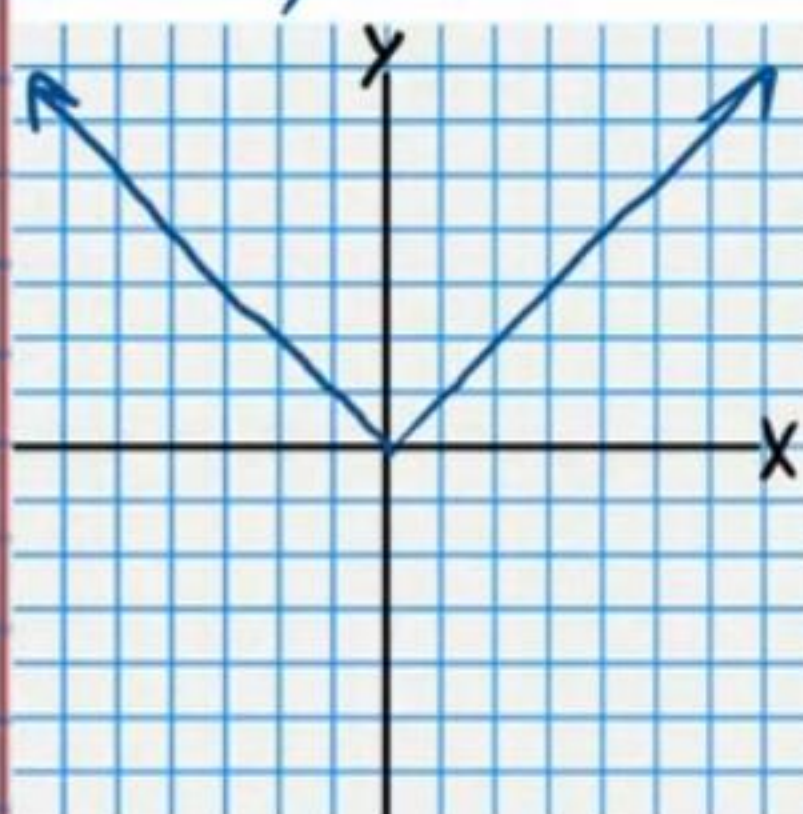
⑩  $y = -|x-3|+1$

Change the form of the equation.

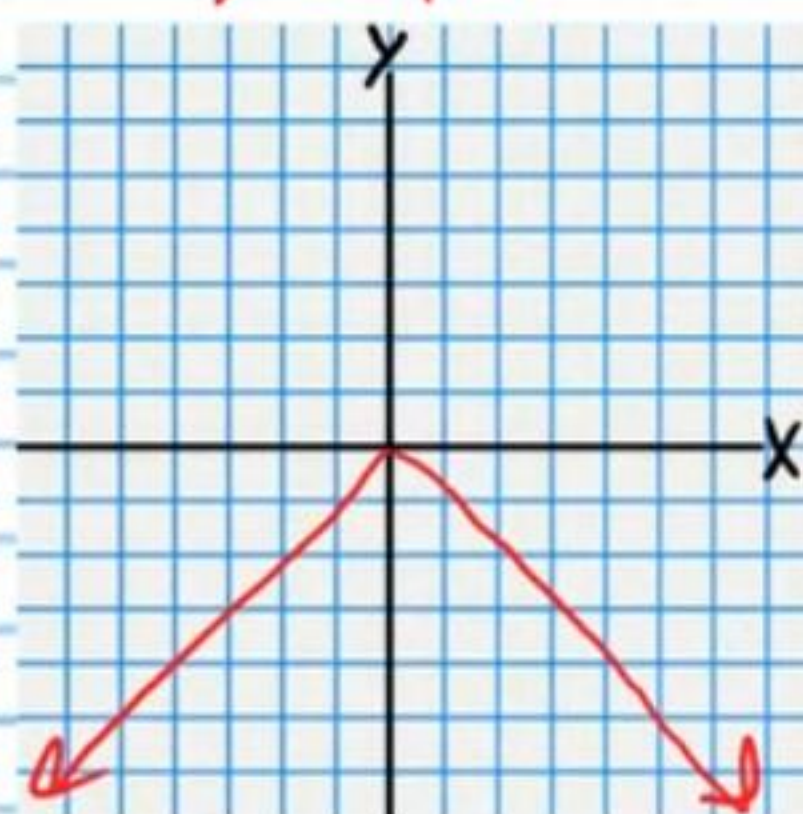
$$\begin{aligned} \Rightarrow y &= -|x-3|+1 \\ y-1 &= -|x-3|+1-1 \\ y-1 &= -|x-3| \\ \frac{y-1}{-1} &= \frac{-|x-3|}{-1} \end{aligned}$$

Altered Form:  $-(y-1) = |x-3|$

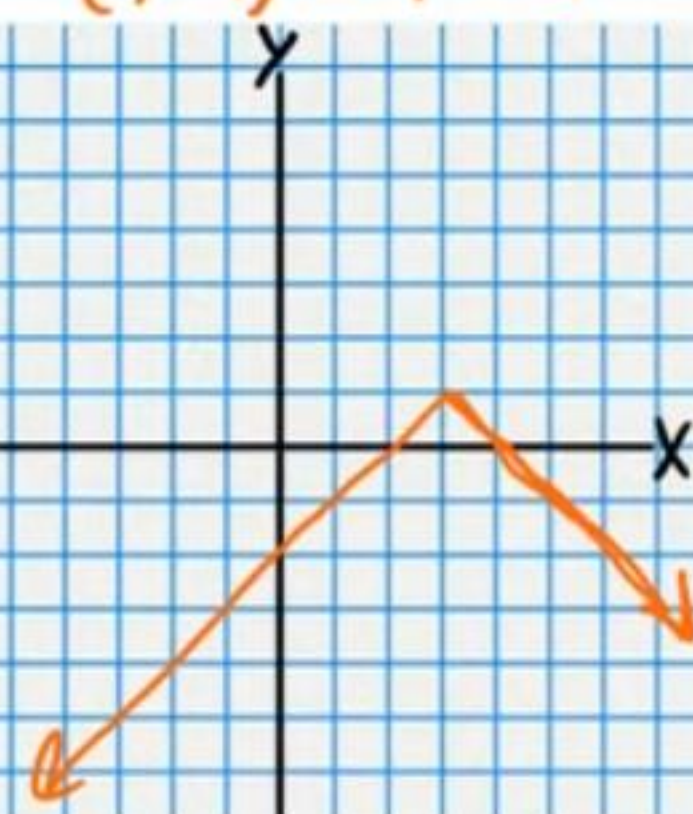
Parent Relation  
 $y = |x|$



Transformation #1  
 $-y = |x|$



Transformation #2  
 $-(y-1) = |x-3|$



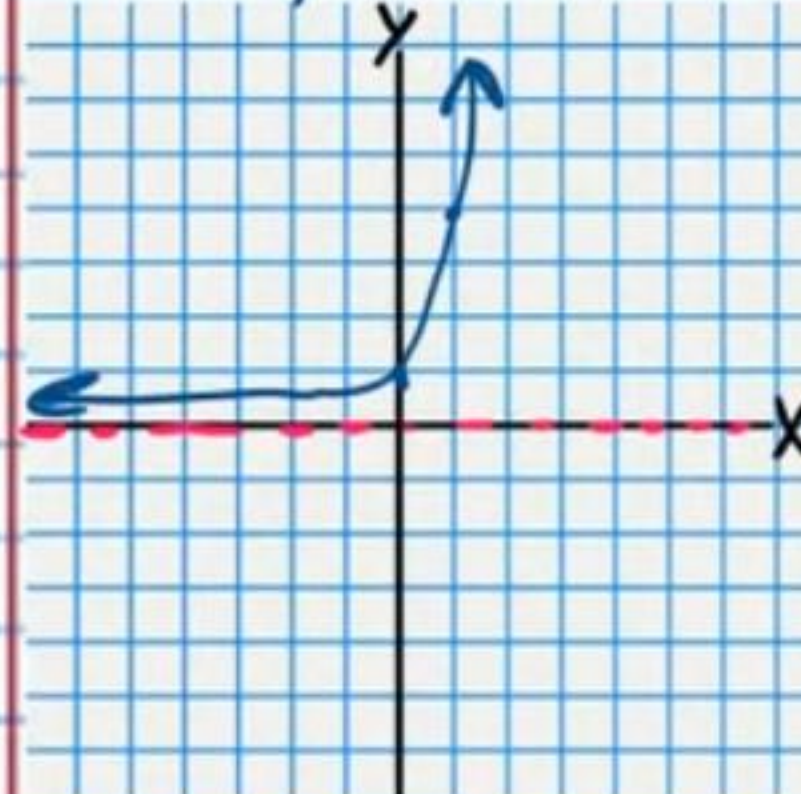
⑪  $y = 3(4^{x+3}) - 5$

Change the form of the equation.

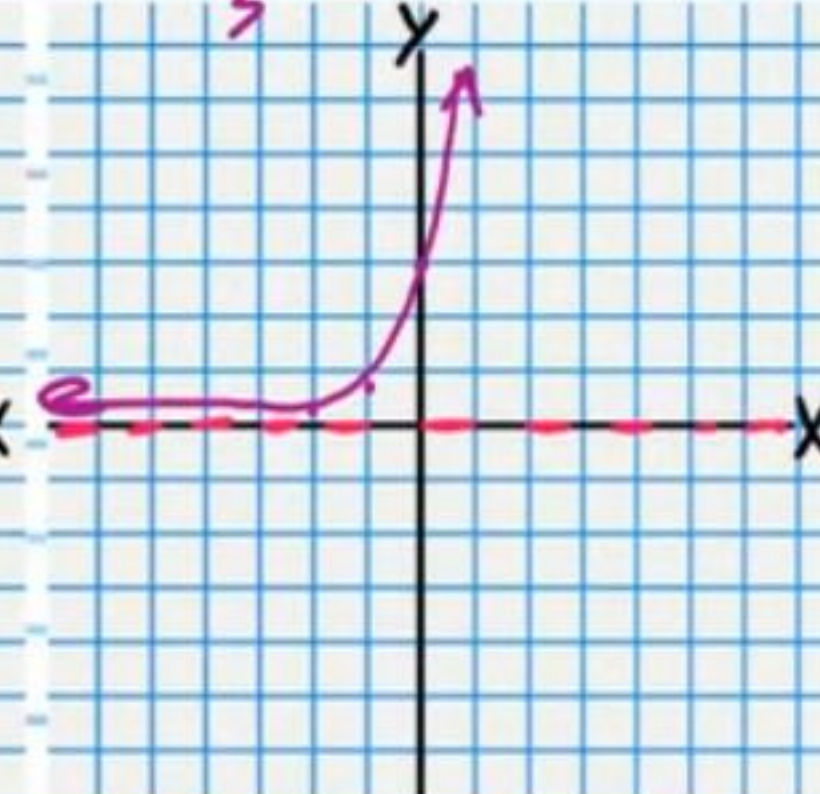
$$\begin{aligned} \Rightarrow y &= 3(4^{x+3}) - 5 \\ y+5 &= 3(4^{x+3}) - 5 + 5 \\ y+5 &= 3(4^{x+3}) \\ \frac{y+5}{3} &= \frac{3(4^{x+3})}{3} \end{aligned}$$

Altered Form:  $\frac{y+5}{3} = 4^{x+3}$

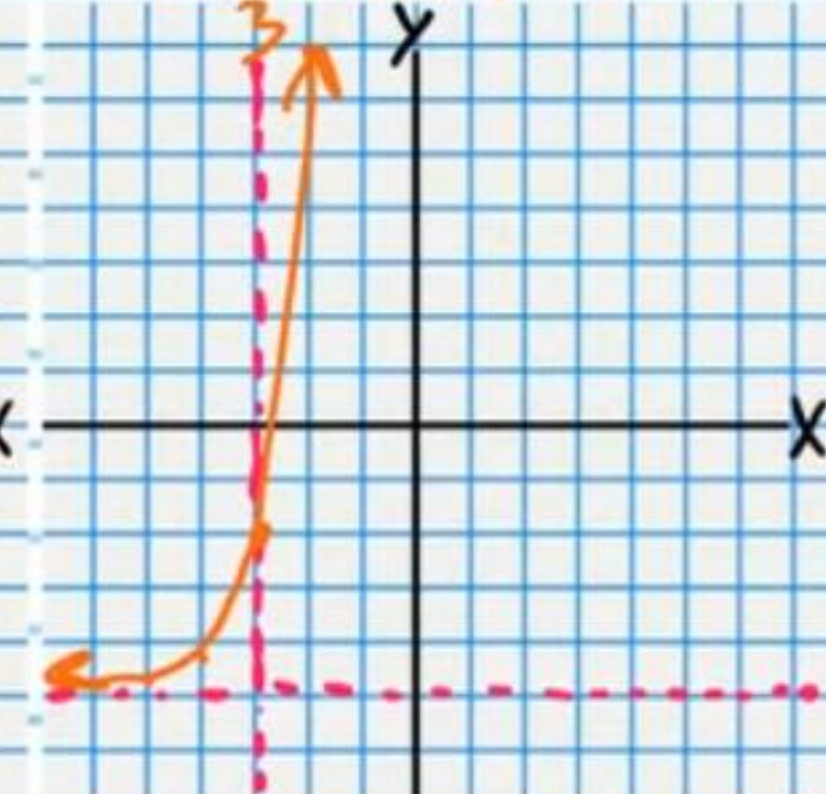
Parent Relation  
 $y = 4^x$



Transformation #1  
 $\frac{y}{3} = 4^x$



Transformation #2  
 $\frac{y+5}{3} = 4^{x+3}$





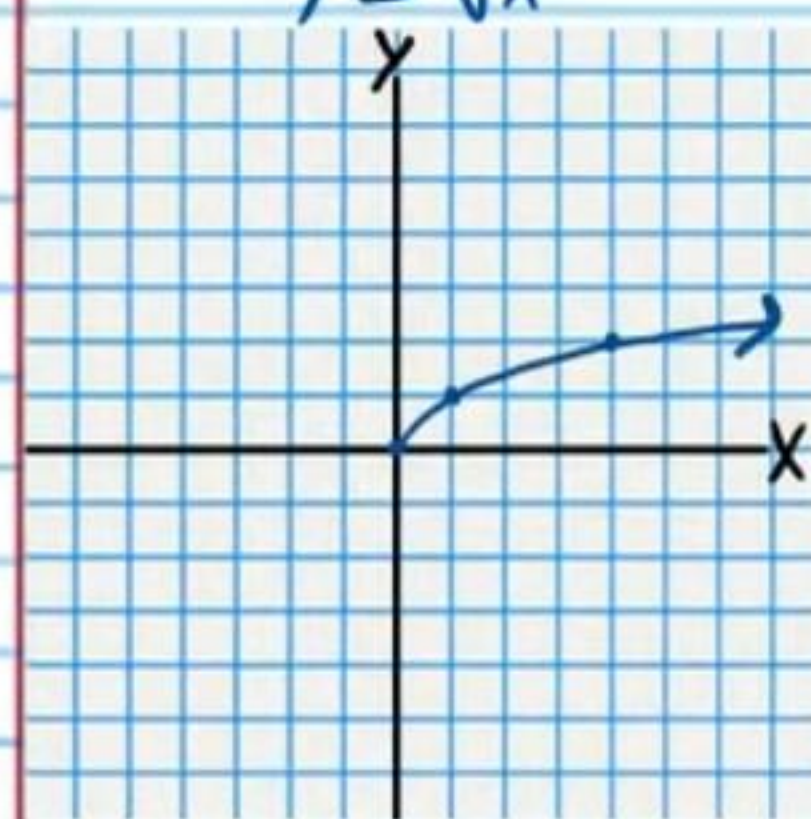
$$\textcircled{12} y = -\sqrt{x+1} + 2$$

Change the form of the equation.

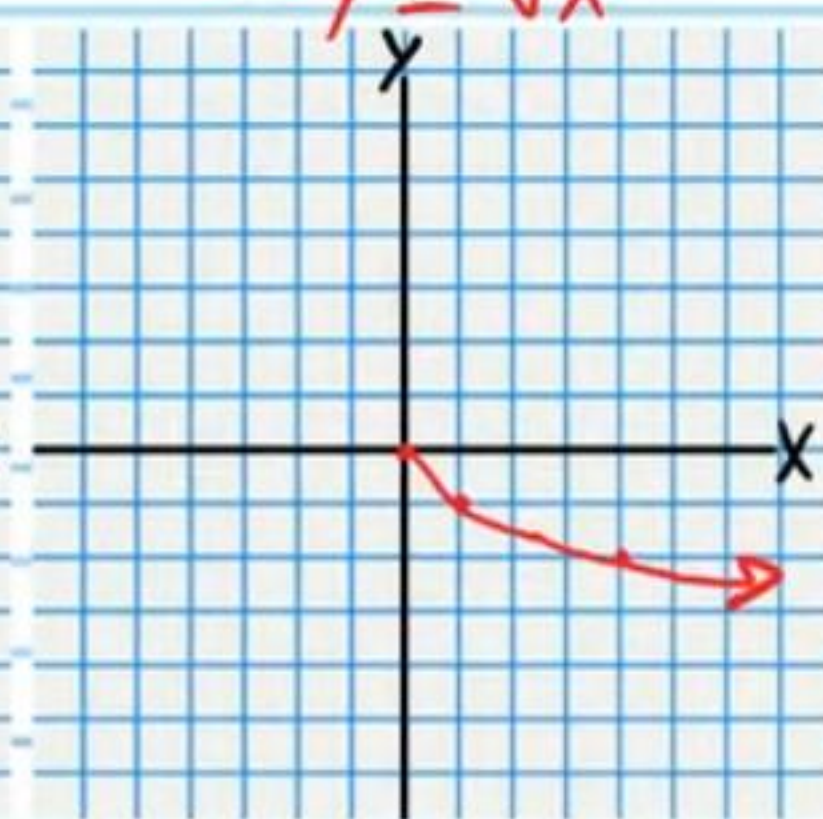
$$\begin{aligned} \Rightarrow y &= -\sqrt{x+1} + 2 \\ y - 2 &= -\sqrt{x+1} + 2 - 2 \\ y - 2 &= -\sqrt{x+1} \\ \frac{y-2}{-1} &= \frac{-\sqrt{x+1}}{-1} \end{aligned}$$

Altered Form:  $-(y-2) = \sqrt{x+1}$

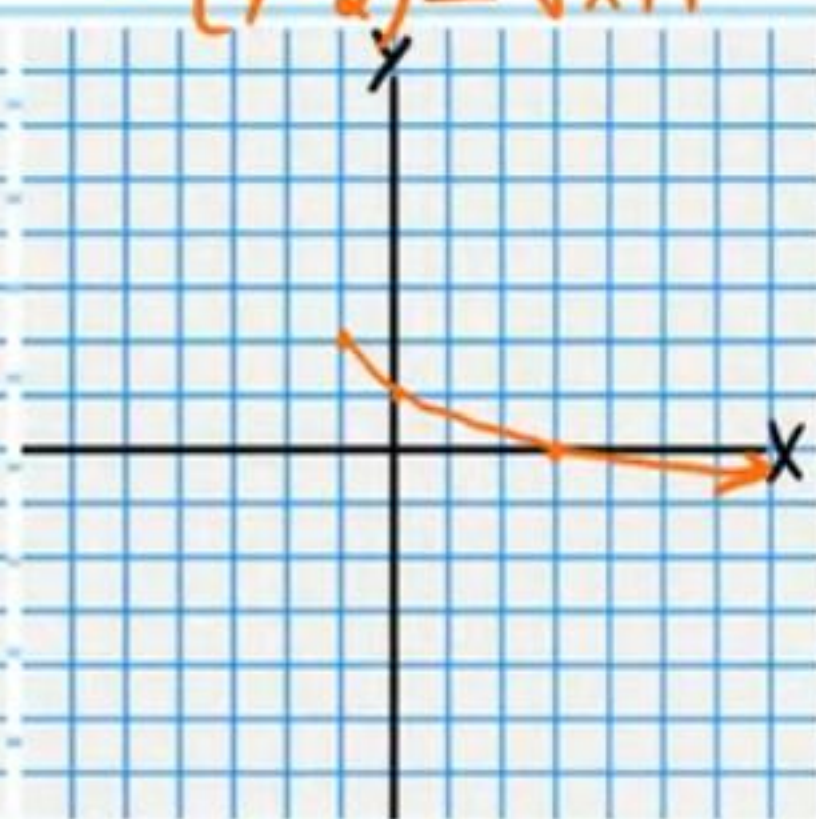
Parent Relation  
 $y = \sqrt{x}$



Transformation #1  
 $-y = \sqrt{x}$



Transformation #2  
 $-(y-2) = \sqrt{x+1}$



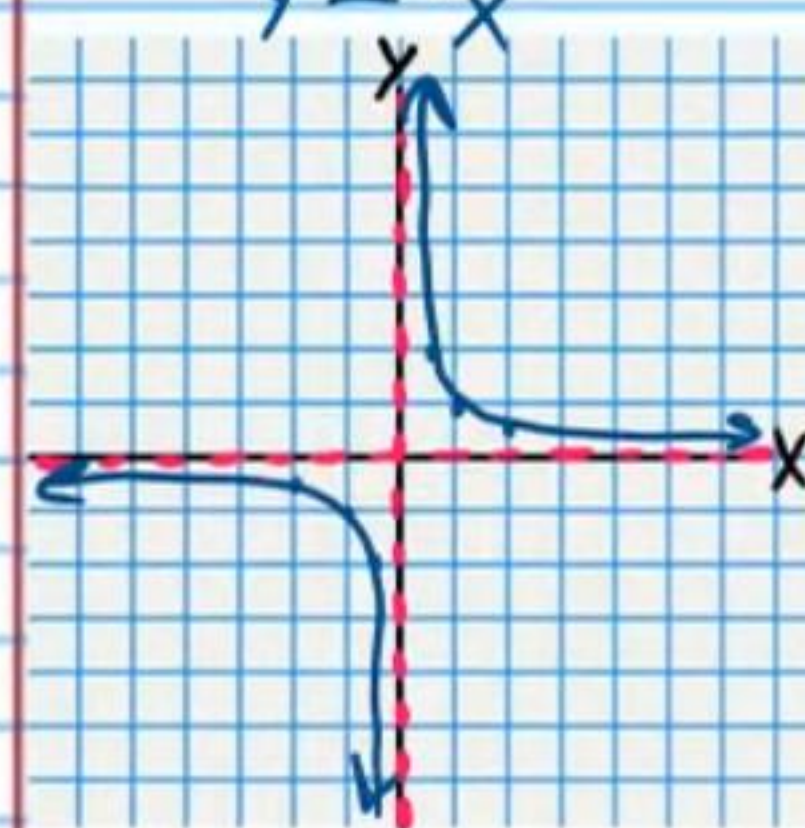
$$\textcircled{13} y = \frac{4}{x+2} - 3$$

Change the form of the equation.

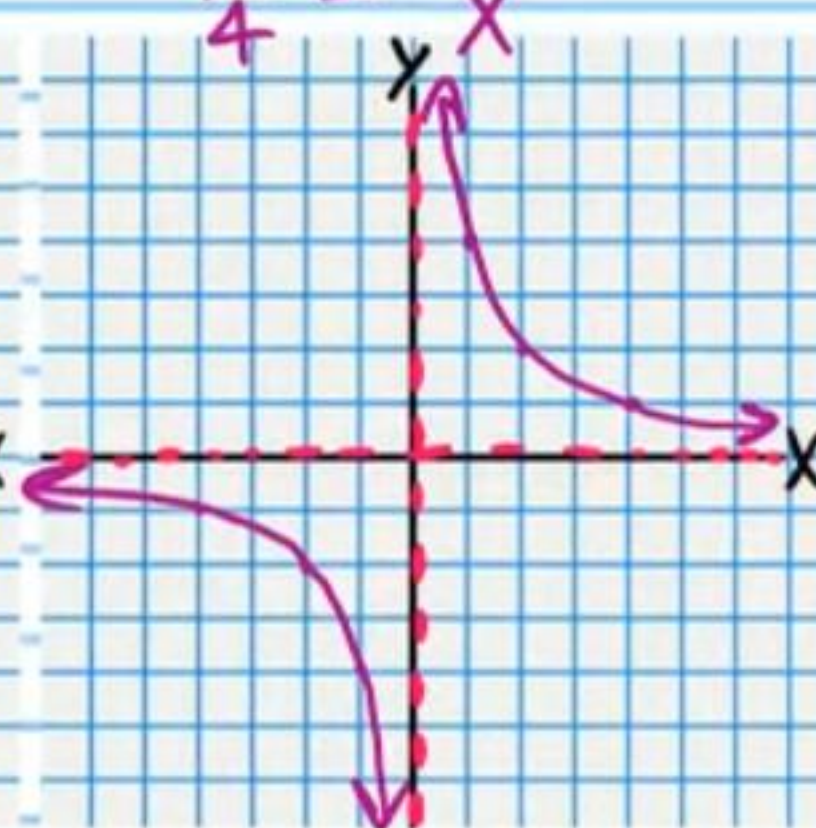
$$\begin{aligned} \Rightarrow y &= \frac{4}{x+2} - 3 \\ y + 3 &= \frac{4}{x+2} - 3 + 3 \\ y + 3 &= \frac{4}{x+2} \\ \frac{1}{4}(y+3) &= \frac{1}{x+2} \left(\frac{1}{4}\right) \end{aligned}$$

Altered Form:  $\frac{y+3}{4} = \frac{1}{x+2}$

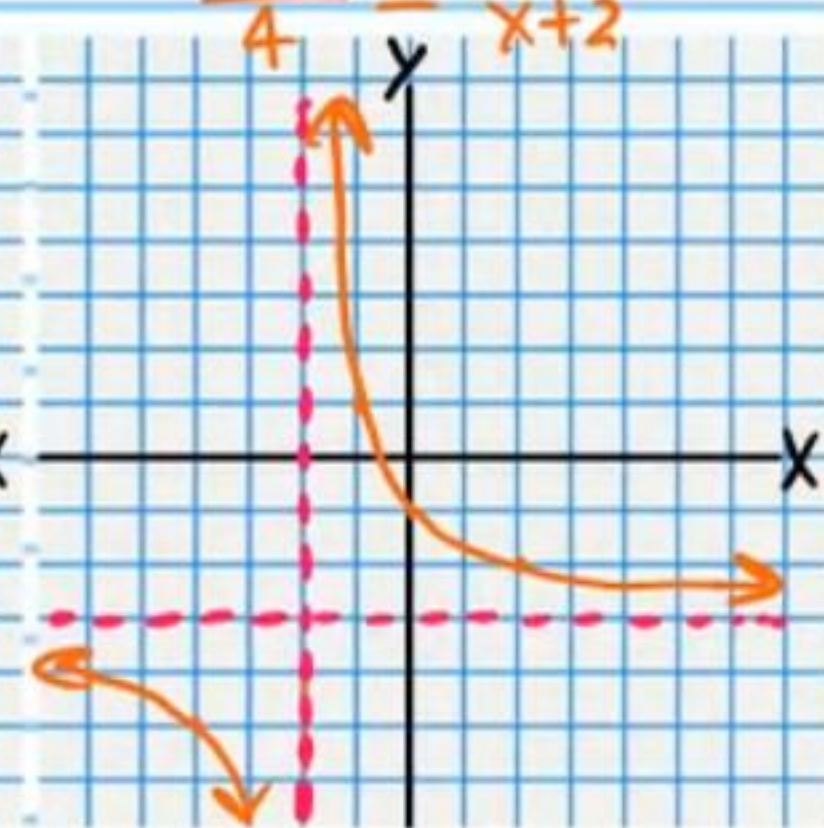
Parent Relation  
 $y = \frac{1}{x}$



Transformation #1  
 $\frac{y}{4} = \frac{1}{x}$



Transformation #2  
 $\frac{y+3}{4} = \frac{1}{x+2}$





$$\textcircled{14} y = \frac{3}{x-1} + 4$$

Change the form of the equation.

$$\Rightarrow y = \frac{3}{x-1} + 4$$

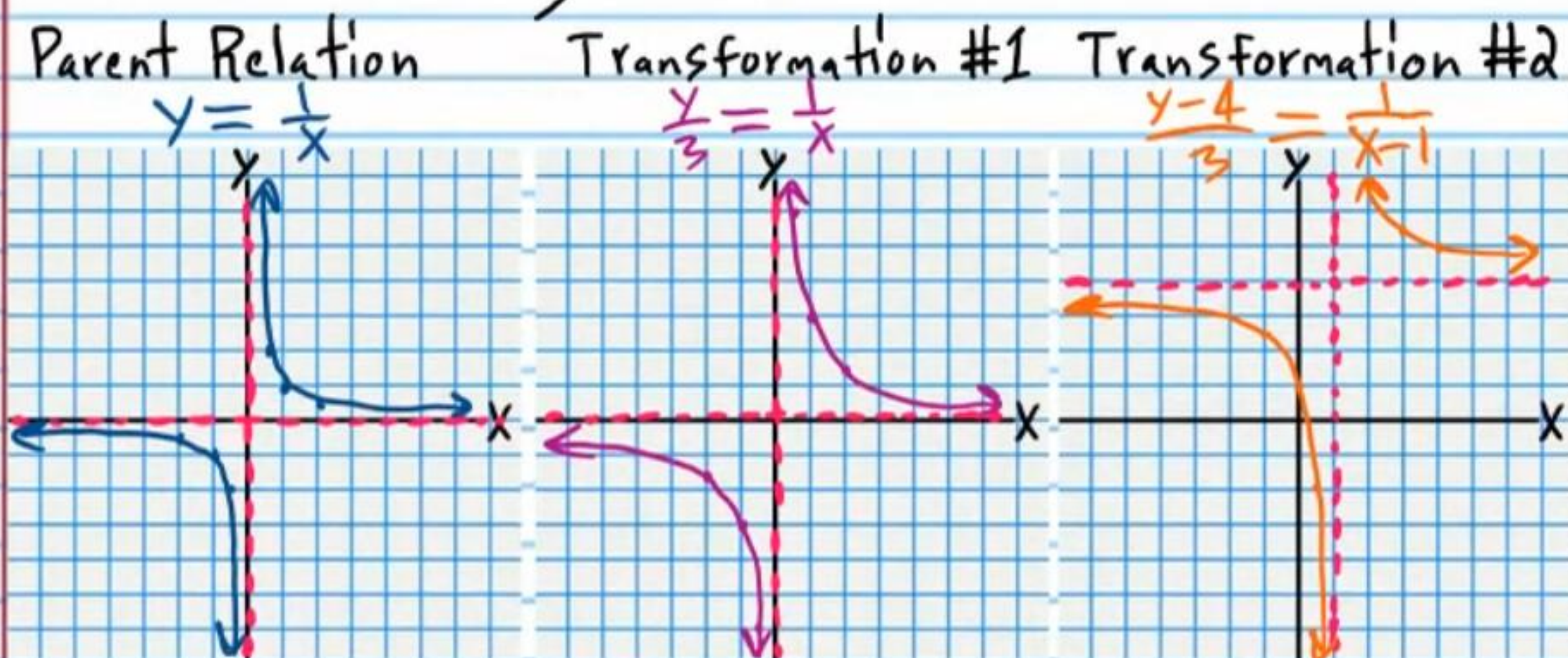
$$y - 4 = \frac{3}{x-1} + 4 - 4$$

$$y - 4 = \frac{3}{x-1}$$

$$\frac{1}{3}(y-4) = \frac{3}{x-1} \left(\frac{1}{3}\right)$$

$$\text{Altered Form: } \frac{y-4}{3} = \frac{1}{x-1}$$

Parent Relation



$$\textcircled{15} y = 2\sqrt{-x-1} - 4$$

Change the form of the equation.

$$\Rightarrow y = 2\sqrt{-x-1} - 4$$

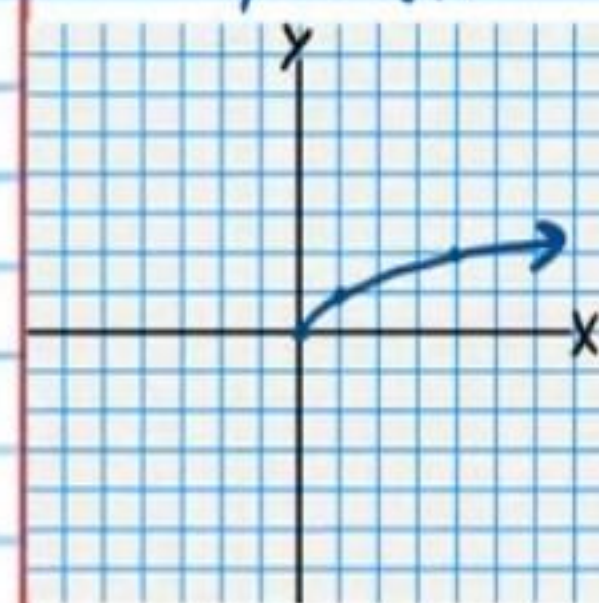
$$y + 4 = 2\sqrt{-x-1} - 4 + 4$$

$$y + 4 = 2\sqrt{-x-1}$$

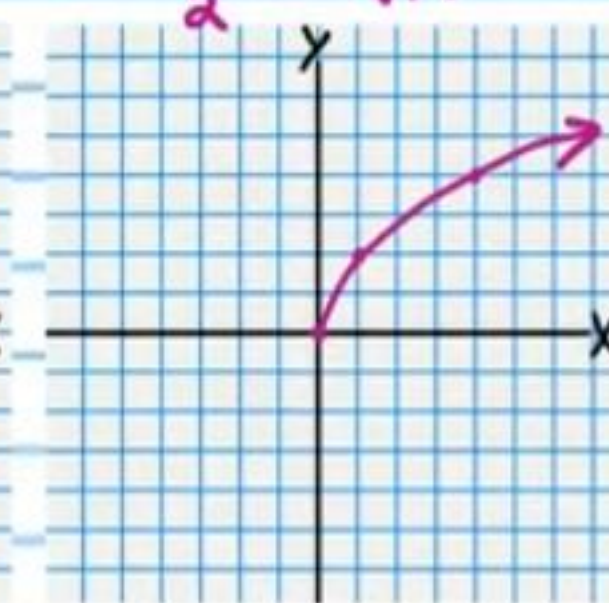
$$\frac{y+4}{2} = \frac{2\sqrt{-x-1}}{2}$$

$$\text{Altered Form: } \frac{y+4}{2} = \sqrt{-x-1}$$

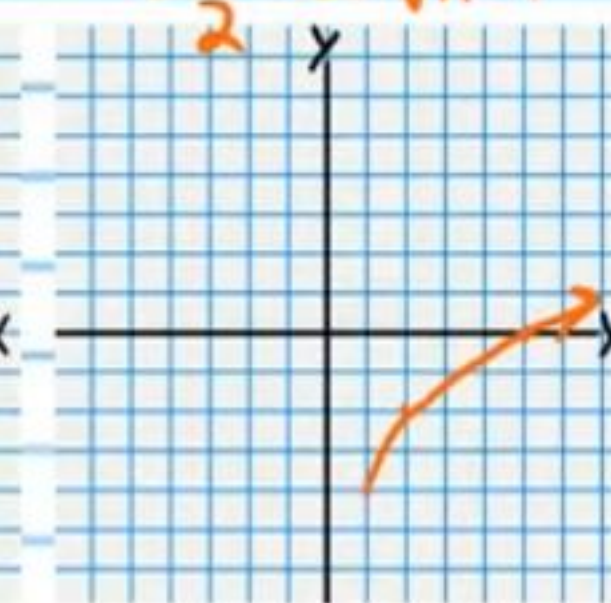
Parent Relation  
 $y = \sqrt{x}$



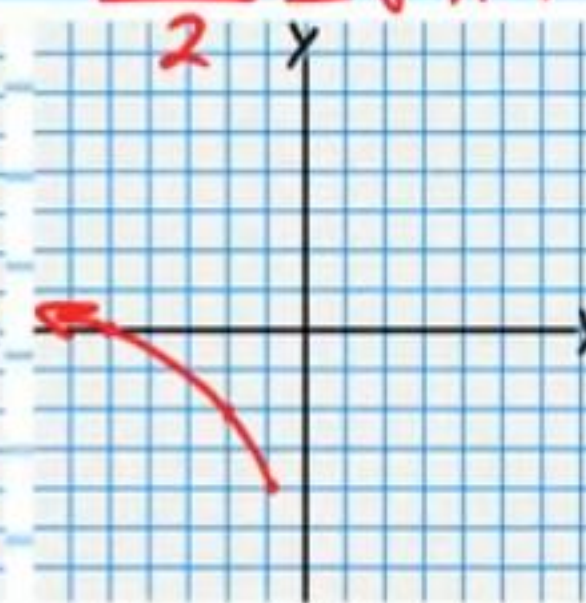
Transformation #1  
 $\frac{y}{2} = \sqrt{x}$



Transformation #2  
 $\frac{y+4}{2} = \sqrt{x-1}$



Final Transformation  
 $\frac{y+4}{2} = \sqrt{-x-1}$



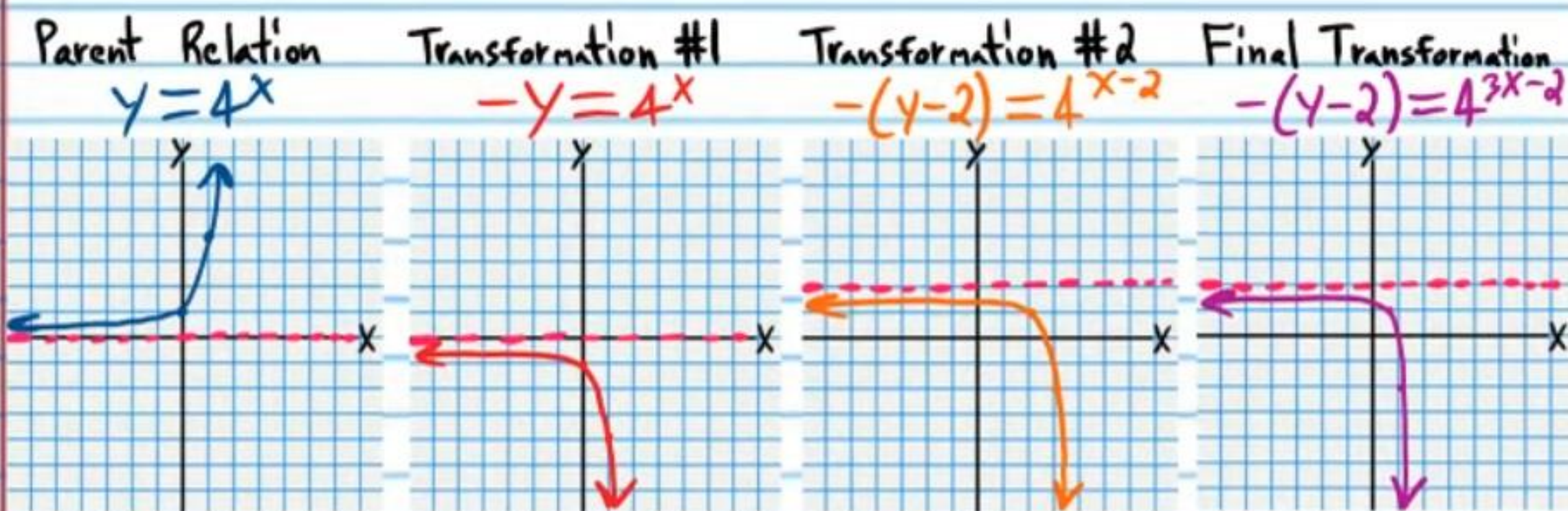


$$\textcircled{16} y = -4^{3x-2} + 2$$

Change the form of the equation.

$$\begin{aligned} \Rightarrow y &= -4^{3x-2} + 2 \\ y - 2 &= -4^{3x-2} + 2 - 2 \\ y - 2 &= -4^{3x-2} \\ \underline{y-2} &= \underline{-4^{3x-2}} \\ -1 & \quad -1 \end{aligned}$$

Altered Form:  $-(y-2) = 4^{3x-2}$

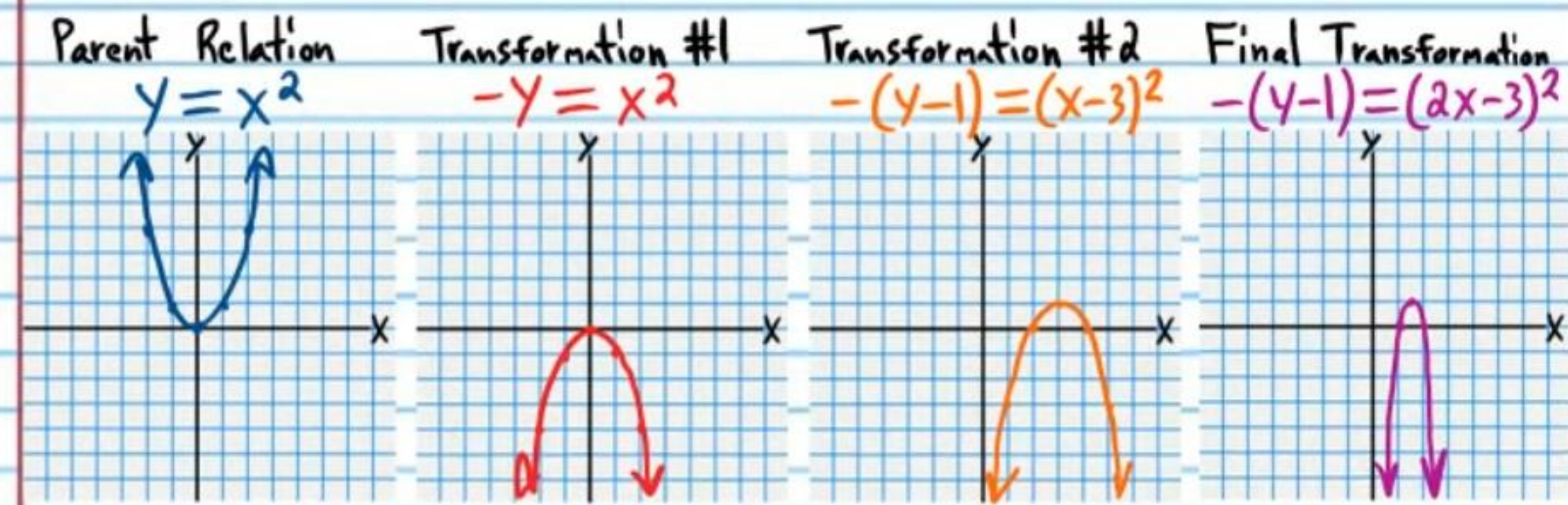


$$\textcircled{17} y = -(2x-3)^2 + 1$$

Change the form of the equation.

$$\begin{aligned} \Rightarrow y &= -(2x-3)^2 + 1 \\ y - 1 &= -(2x-3)^2 + 1 - 1 \\ y - 1 &= -(2x-3)^2 \\ \underline{y-1} &= \underline{-(2x-3)^2} \\ -1 & \quad -1 \end{aligned}$$

Altered Form:  $-(y-1) = (2x-3)^2$





$$\textcircled{18} \ y = \frac{\log_3(-x+2)}{3} - 4$$

Change the form of the equation.

$$\Rightarrow y = \frac{\log_3(-x+2)}{3} - 4$$

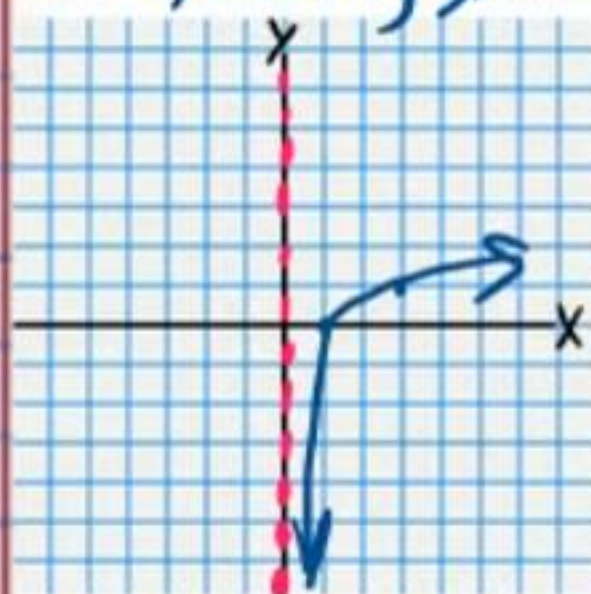
$$y+4 = \frac{\log_3(-x+2)}{3} - 4 + 4$$

$$y+4 = \frac{\log_3(-x+2)}{3}$$

$$3(y+4) = \frac{\log_3(-x+2)}{3} \quad (3)$$

Altered Form:  $3(y+4) = \log_3(-x+2)$

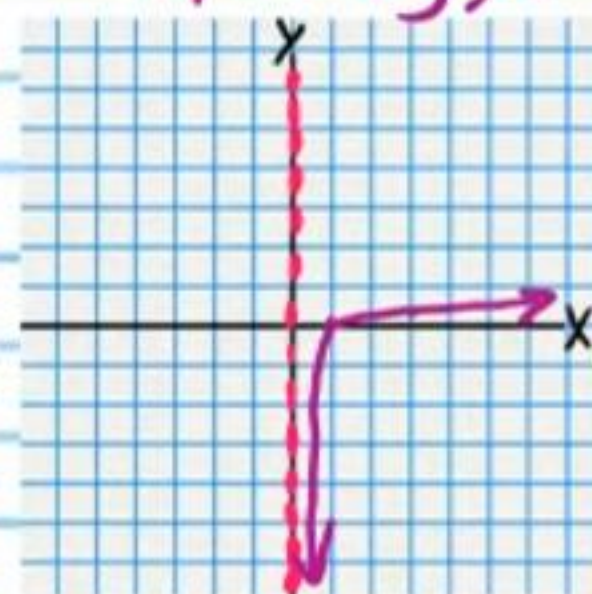
Parent Relation  
 $y = \log_3 x$



Logarithm Property III

$$\log_b b = 1$$

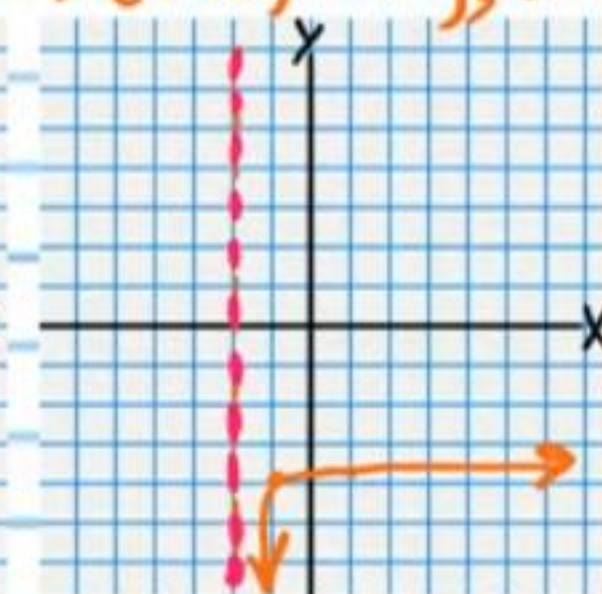
Transformation #1  
 $3y = \log_3 x$



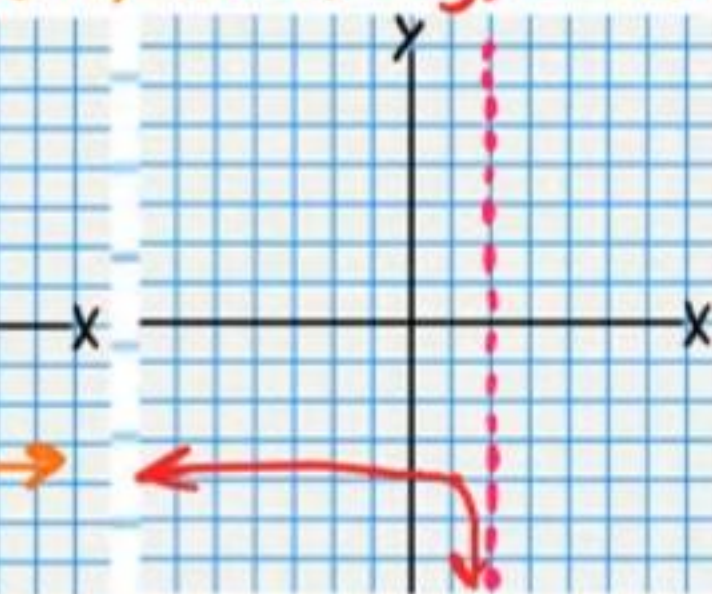
Logarithm Property II

$$\log_b b^x = x$$

Transformation #2  
 $3(y+4) = \log_3(x+2)$



Final Transformation  
 $3(y+4) = \log_3(-x+2)$



Logarithm Property IX

$$\log_b 1 = 0$$

$$\textcircled{19} \ y = -\frac{(x-2)^3}{2} - 1$$

Change the form of the equation.

$$\Rightarrow y = -\frac{(x-2)^3}{2} - 1$$

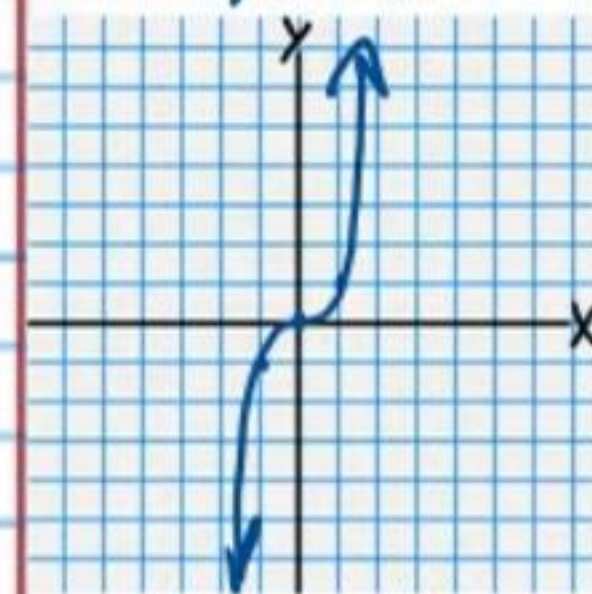
$$y+1 = -\frac{(x-2)^3}{2} - 1 + 1$$

$$y+1 = -\frac{(x-2)^3}{2}$$

$$-2(y+1) = -\frac{(x-2)^3}{2} (-2)$$

Altered Form:  $-2(y+1) = (x-2)^3$

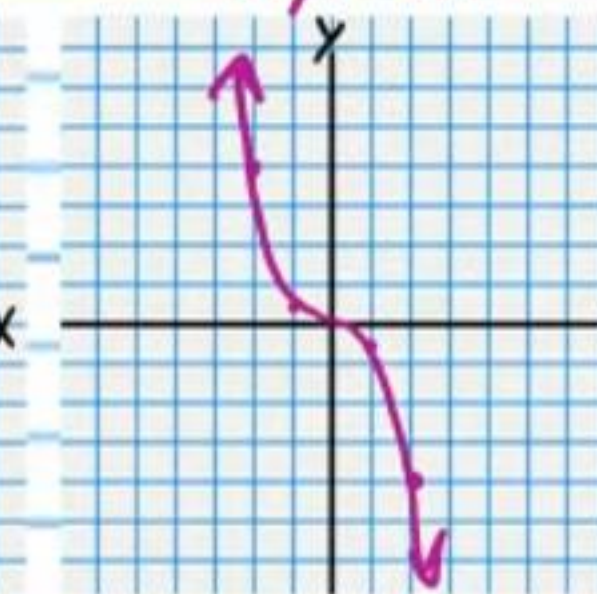
Parent Relation  
 $y = x^3$



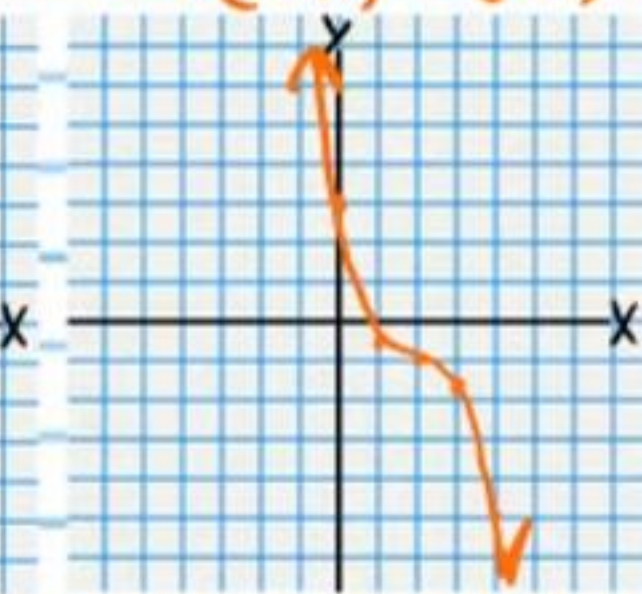
Transformation #1  
 $-y = x^3$



Transformation #2  
 $-2y = x^3$



Final Transformation  
 $-2(y+1) = (x-2)^3$



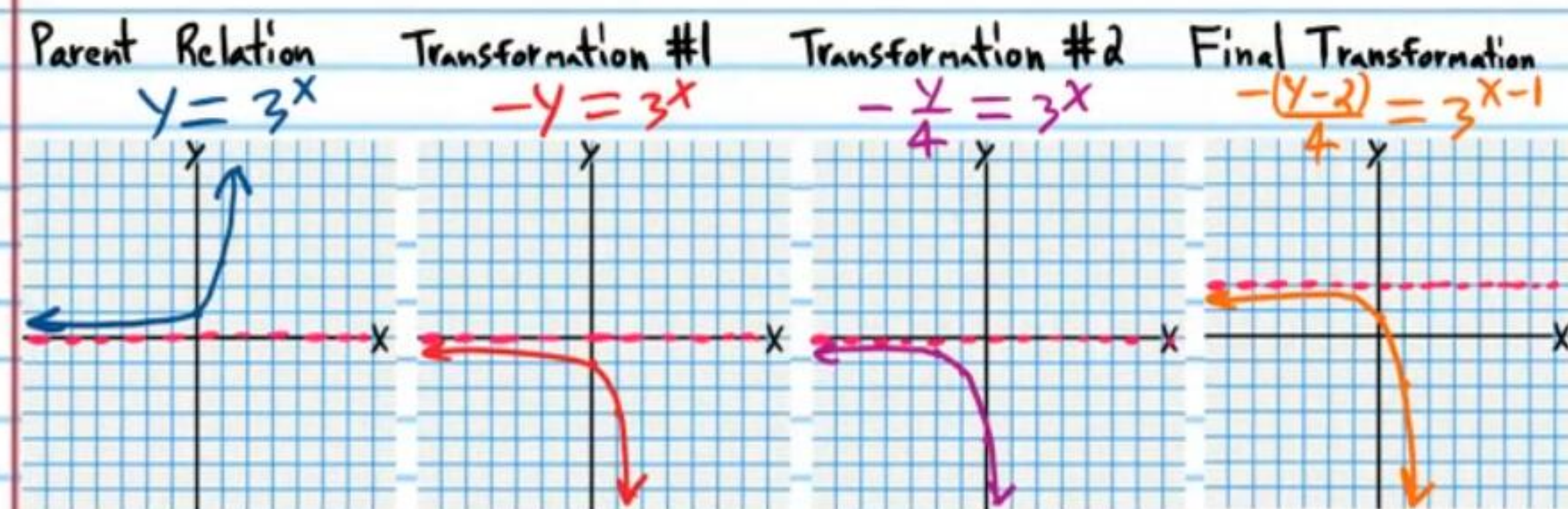


②①  $y = -4(3^{x-1}) + 2$

Change the form of the equation.

$$\begin{aligned} \Rightarrow y &= -4(3^{x-1}) + 2 \\ y - 2 &= -4(3^{x-1}) + 2 - 2 \\ y - 2 &= -4(3^{x-1}) \\ \frac{y-2}{-4} &= \frac{-4(3^{x-1})}{-4} \end{aligned}$$

Altered Form:  $\frac{-(y-2)}{4} = 3^{x-1}$

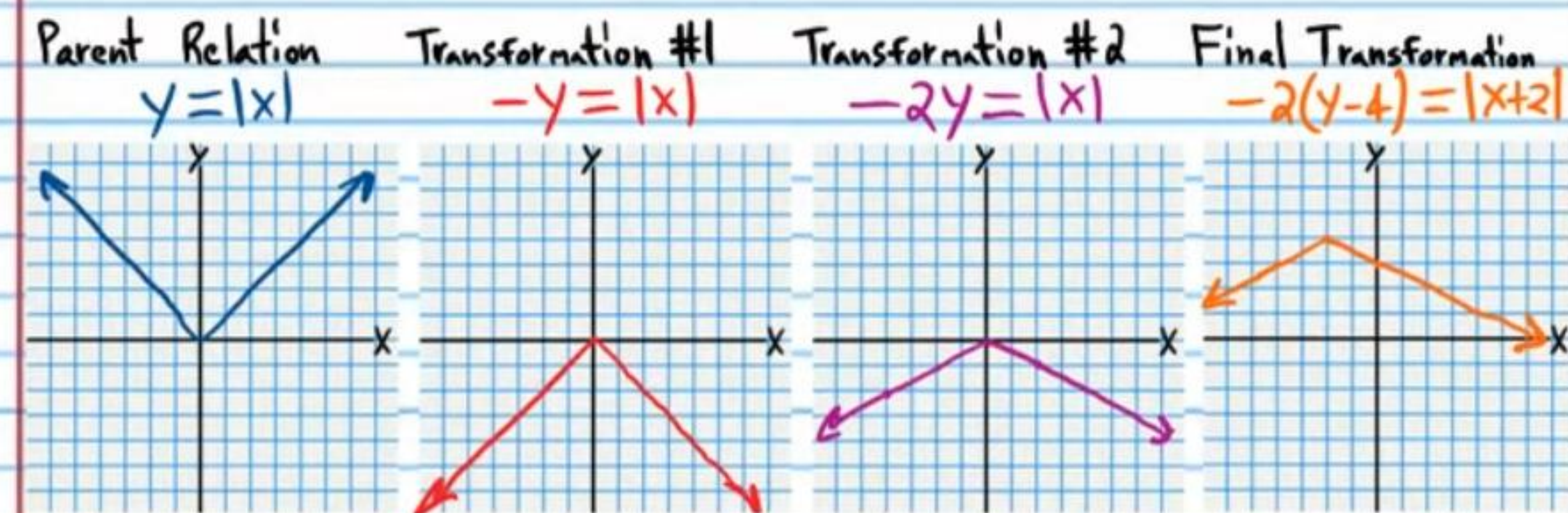


②①  $y = -\frac{|x+2|}{2} + 4$

Change the form of the equation.

$$\begin{aligned} \Rightarrow y &= -\frac{|x+2|}{2} + 4 \\ y - 4 &= -\frac{|x+2|}{2} + 4 - 4 \\ y - 4 &= -\frac{|x+2|}{2} \\ -2(y-4) &= -\frac{|x+2|}{2} (-2) \end{aligned}$$

Altered Form:  $-2(y-4) = |x+2|$





②  $y = -3\log_2(x-2) - 2$

Change the form of the equation.

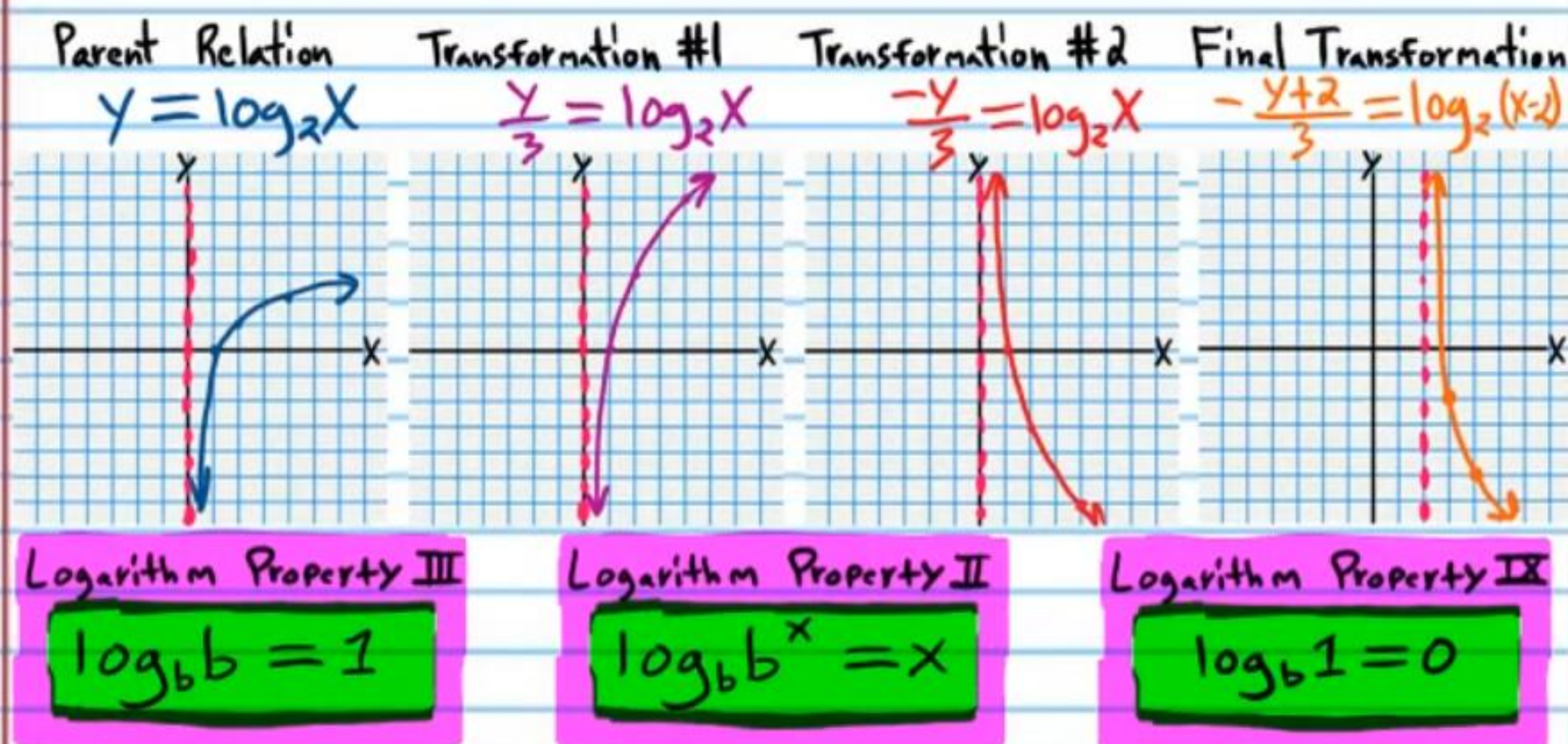
$\Rightarrow y = -3\log_2(x-2) - 2$

$$y+2 = -3\log_2(x-2) - 2 + 2$$

$$y+2 = -3\log_2(x-2)$$

$$\frac{y+2}{-3} = \frac{-3\log_2(x-2)}{-3}$$

Altered Form:  $-\frac{y+2}{3} = \log_2(x-2)$



③  $y = 1 - \sqrt{\frac{x}{2} - 1}$

Change the form of the equation.

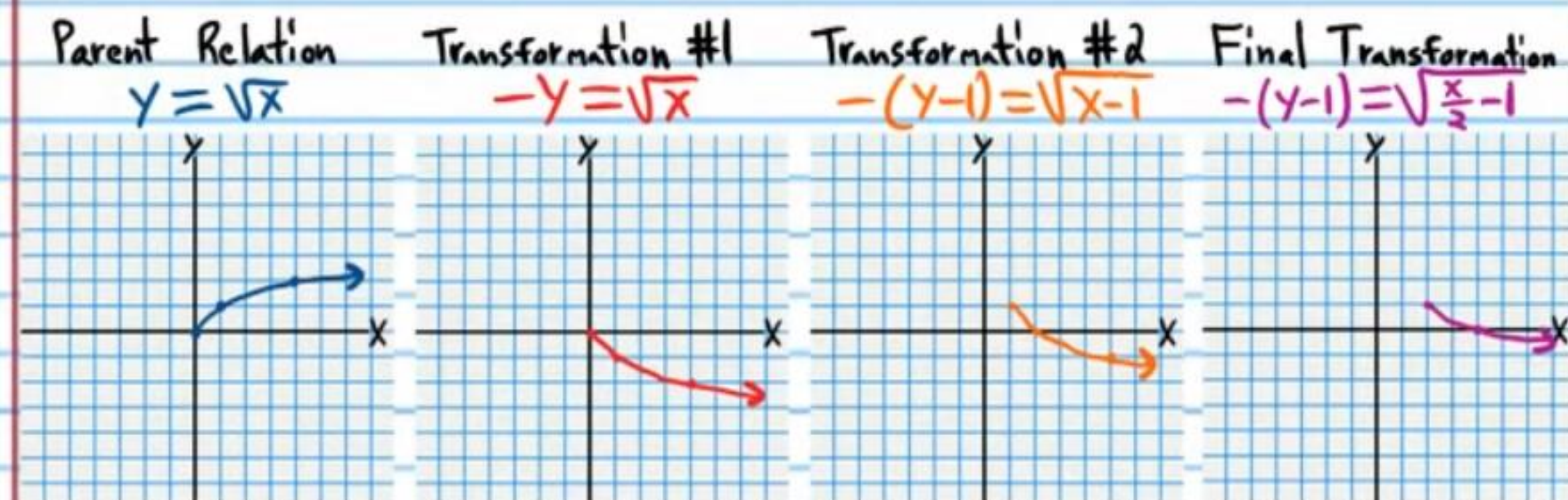
$\Rightarrow y = 1 - \sqrt{\frac{x}{2} - 1}$

$$y-1 = 1 - \sqrt{\frac{x}{2} - 1} - 1$$

$$y-1 = -\sqrt{\frac{x}{2} - 1}$$

$$\frac{y-1}{-1} = \frac{-\sqrt{\frac{x}{2} - 1}}{-1}$$

Altered Form:  $-(y-1) = \sqrt{\frac{x}{2} - 1}$





②④  $y = -3 - \frac{|x-4|}{3}$

Change the form of the equation.

$\Rightarrow y = -3 - \frac{|x-4|}{3}$

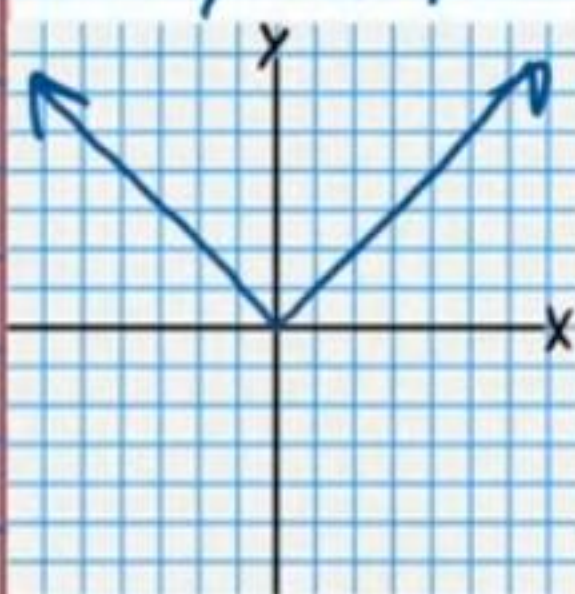
$y+3 = -3 - \frac{|x-4|}{3} + 3$

$y+3 = -\frac{|x-4|}{3}$

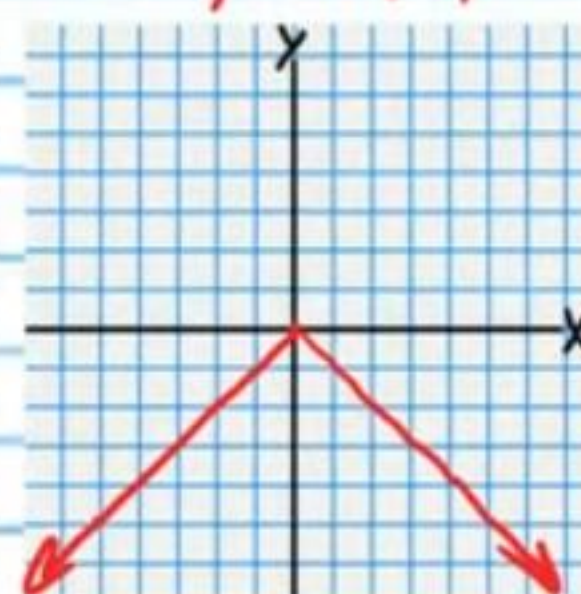
$-3(y+3) = -\frac{|x-4|}{3} (-3)$

Altered Form:  $-3(y+3) = |x-4|$

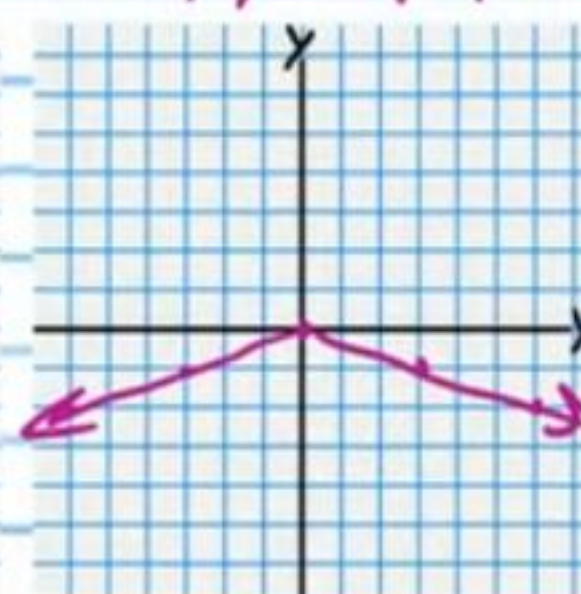
Parent Relation  
 $y = |x|$



Transformation #1  
 $-y = |x|$



Transformation #2  
 $-3y = |x|$



Final Transformation  
 $-3(y+3) = |x-4|$



②⑤  $y = -\frac{2}{x-1} + 4$

Change the form of the equation.

$\Rightarrow y = -\frac{2}{x-1} + 4$

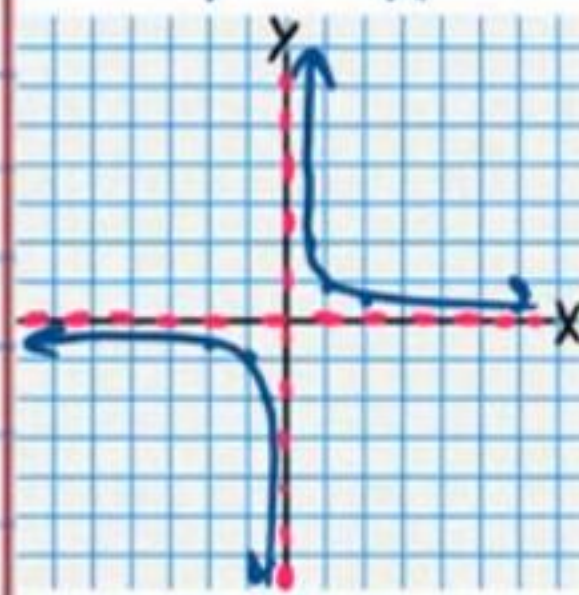
$y-4 = -\frac{2}{x-1} + 4 - 4$

$y-4 = -\frac{2}{x-1}$

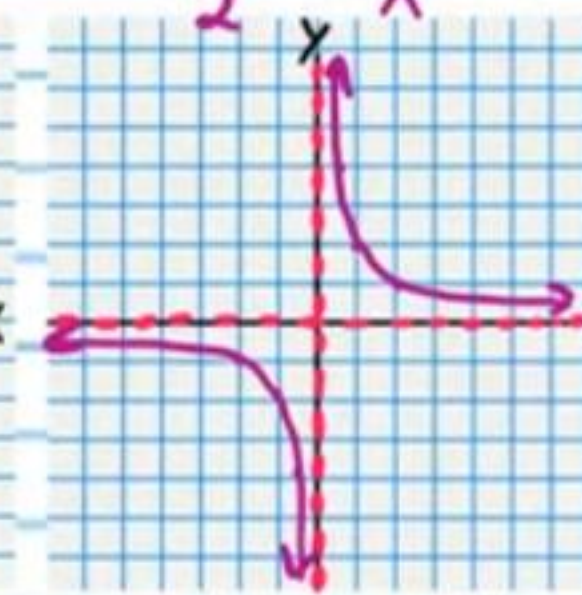
$-\frac{1}{2}(y-4) = -\frac{2}{x-1} (-\frac{1}{2})$

Altered Form:  $-\frac{(y-4)}{2} = \frac{1}{x-1}$

Parent Relation  
 $y = \frac{1}{x}$



Transformation #1  
 $\frac{y}{2} = \frac{1}{x}$



Transformation #2  
 $-\frac{y}{2} = \frac{1}{x}$



Final Transformation  
 $-\frac{(y-4)}{2} = \frac{1}{x-1}$





②⑥  $y = -\frac{3}{x+2} - 1$

Change the form of the equation.

$\Rightarrow y = -\frac{3}{x+2} - 1$

$$y+1 = -\frac{3}{x+2} - 1+1$$

$$y+1 = -\frac{3}{x+2}$$

$$-\frac{1}{3}(y+1) = -\frac{3}{x+2} \left(-\frac{1}{3}\right)$$

Altered Form:  $-\frac{(y+1)}{3} = \frac{1}{x+2}$

